

Group Theory

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Contents

1	Introduction	4
1.1	Basic Definitions and Examples	4
1.2	$\mathbb{Z}_m$ and $\mathbb{Z}_m^*$	5
1.3	Exercises: Group Verification	6
1.4	Group Properties	9
1.4.1	Identity	9
1.4.2	Inverse	9
1.4.3	Isomorphisms	10
1.4.4	Order	11
2	Symmetric Groups	13
2.1	Definition	13
2.2	Cycle Form	13
2.3	k -Cycles	15
2.4	Order	17
2.5	Fun: Counting k -cycles	19
3	Dihedral Groups	22
3.1	Definition	22
3.2	Order	23
4	The Lagrange's Theorem	24
4.1	Subgroups	24
4.1.1	Definition	24
4.1.2	Subgroup Tests	24
4.2	Cyclic Groups	27
4.2.1	Definition	27
4.2.2	Cyclic Subgroups	28
4.2.3	Double Generators	33
4.3	Cosets	34
4.3.1	Definition	34
4.3.2	Properties of Cosets	35
4.4	Proof of Lagrange's Theorem	37
5	Alternating Groups	39
5.1	Definition	39
5.2	Properties	40

6	Homomorphisms	42
6.1	Definition	42
6.2	Kernel and Image	43
6.3	Isomorphisms (Revised)	45
6.3.1	Definition	45
6.4	The First Isomorphism Theorem	46
6.5	Classification of Finite Abelian Subgroups	48
7	Quotient Groups	49
7.1	Definition	49
7.2	Towards The Correspondence Theorem	53
7.3	The Correspondence Theorem	57
8	Group Actions	60
8.1	Definition	60
8.2	Burnside's Lemma	62
A	Number Theory	64
A.1	Divisibility	64
A.2	Greatest Common Divisors	64
A.3	Linear Diophantine Equations	65
B	Mapping	66
B.1	Surjective and Injective Functions	66
B.2	Bijections	66
C	Indices	70

List of Tables

1.4.1 (A) $\{\pm 1\}$ and (B) $\mathbb{Z}_2$	10
1.4.2 Multiplication Table of Order-2 Groups	10
1.4.3 Multiplication Table of Order-3 Groups	10

List of Figures

7.1 Ancestor trees	57
7.2 Partition Tree of $\mathbb{Z}_8$	59

1 Introduction

1.1 Basic Definitions and Examples

To have a clear grasp of what a group looks like, consider the set of integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ and the addition operator $+$ applied on elements in $\mathbb{Z}$. We have that

- $\mathbb{Z}$ is closed under addition since the sum of two integers is still an integer.
- $\mathbb{Z}$ is associative under addition, e.g. $(1 + 2) + 3 = 1 + (2 + 3) = 6$.
- The identity element 0 plus any integer is the integer itself, e.g. $0 + 1 = 1, 0 + 2 = 2$, etc.
- An integer has an reverse where their sum is 0, e.g. $1 + (-1) = 0$.

These are the four principle properties of an additive group. As an example of invalid group, consider $\mathbb{Z}$ with multiplication $\times$. Clearly the identity is 1 since any integer times 1 is itself, but unfortunately 0 will have no inverse as $\frac{1}{0}$ is undefined. $(\mathbb{Z}, \times)$ is thus not a group.

Definition 1.1.1 (Groups)

A group is a set G with an operation $*$: $G \times G \mapsto G$, denoted $(G, *)$, that satisfies

1. $\forall g, h \in G, \quad g * h \in H$. (Closure of Operation)
2. $\forall g, h, k \in G, \quad (g * h) * k = g * (h * k)$. (Associativity)
3. $\exists e \in G$ such that $\forall g \in G, \quad e * g = g * e = g$. (Existence of Identity)
4. $\forall g \in G, \quad \exists g^{-1} \in G$ such that $g * g^{-1} = g^{-1} * g = e$. (Invertibility)

Now look at an example. We define $\mathbb{R}^{n \times n}$ as the set of all $n \times n$ real matrices where $n \geq 2$ and the operation is the ordinary matrix multiplication, which is associative. $\mathbb{R}^{n \times n}$ is closed under multiplication since the product of two $n \times n$ matrices is still $n \times n$. The identity is I_n . However, a matrix is invertible if and only its determinant is non-zero. Therefore, quite many matrices in $\mathbb{R}^{n \times n}$ have no inverses; this is not a valid group. But what if we only focus on the invertible matrices? Define the set of general matrices as

$$GL_n(\mathbb{R}) = \{A \in \mathbb{R}^{n \times n} : \det A \neq 0\} \quad (1.1.1)$$

It satisfies the invertibility qualification. Along with the other three properties (operation closure, associativity, identity existence) shown above, $GL_n(\mathbb{R})$ is a valid group.

Here is a more interesting example. Define $\mathbb{Z}^{n \times n}$ as the set of all $n \times n$ integer matrices. Similar to $GL_n(\mathbb{R})$, it satisfies properties 1-3 in 1.1.1. What qualification is necessary if we want $A \in \mathbb{Z}^{n \times n}$ to be invertible? We definitely need $\det A \neq 0$ but we also want A^{-1} to be integral. Note that by property of determinant,

$$\begin{aligned} AA^{-1} = I &\implies \det(AA^{-1}) = \det(I) = 1 \\ &\implies \det(A) \det(A)^{-1} = 1 \end{aligned}$$

Since A, A^{-1} are both integral, we know $\det(A), \det(A)^{-1} \in \mathbb{Z}$. The only case is $|\det(A)| = 1$. So the set of invertible integral matrices can be defined by

$$GL_n(\mathbb{Z}) = \{A \in \mathbb{Z}^{n \times n} : \det A = \pm 1\} \quad (1.1.2)$$

Other example groups are $(\{\pm 1\}, \times), (\{\pm 1, \pm i\}, \times), (\mathbb{R}^n, +)$.

Note that property 3 and 4 in 1.1.1 must be commutative, but the group operation is generally not commutative. For example, the matrix product AB does not necessarily equal BA , where $A, B \in GL_n(\mathbb{R})$. A group with commutative operation is called *abelian*.

Definition 1.1.2 (Abelian Groups)

Let $(G, *)$ be a group. G is abelian if

$$\forall g, h \in G, \quad g * h = h * g$$

For example, $(\mathbb{R}^n, +)$, $(\mathbb{Z}, +)$, $(\{\pm 1\}, \times)$, $(\{\pm 1, \pm i\}, \times)$ are abelian groups since addition and multiplication are commutative operators.

Remark

A subtle but important observation is that the product of two group elements must be unique.

1.2 $\mathbb{Z}_m$ and $\mathbb{Z}_m^*$

We define

$$\mathbb{Z}_m = \{(a \bmod m) : a \in \mathbb{Z}\}, \quad a \cdot b = (a + b \bmod m) \quad (1.2.1)$$

and

$$\mathbb{Z}_m^* = \{a \in \mathbb{Z}_m : a \text{ has an inverse}\}, \quad a \cdot b = (a \times b \bmod m)$$

The definition of $\mathbb{Z}_m^*$ is incomplete. We will elaborate on it. Note that

$$\mathbb{Z}_m = \{0, 1, \dots, m-1\}$$

and $|\mathbb{Z}_m| = m$. The identity in $\mathbb{Z}_m$ is 0 since $(0 + a \bmod m) = a \forall a \in \mathbb{Z}_m$. The inverse of $a \in \mathbb{Z}_m$ is $b \in \mathbb{Z}_m$ satisfying

$$(a + b \bmod m) = (0 \bmod m)$$

For example, consider

$$\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$$

We have $(0 + 0 \bmod 5) = 0$, $(1 + 4 \bmod 5) = 0$, $(2 + 3 \bmod 5) = 0$, so

$$0^{-1} = 0, \quad 1^{-1} = 4, \quad 2^{-1} = 3, \quad 3^{-1} = 2, \quad 4^{-1} = 1$$

$\mathbb{Z}_m^*$ is more complicated. The identity is 1, since $(1 \times a \bmod m) = (a \bmod m) = a$. Similar to $\mathbb{Z}_m$, the inverse of $a \in \mathbb{Z}_m^*$ is $b \in \mathbb{Z}_m^*$ where

$$(a \times b \bmod m) = (1 \bmod m)$$

How do we know which elements in $\{0, 1, \dots, m-1\}$ are invertible, i.e. have an inverse in the same set? For all $a \in \{0, 1, \dots, m-1\}$, consider

$$(a \times b \bmod m) = 1 \implies ab - mk = 1 \text{ for some } k \in \mathbb{Z}$$

By [A.3.2](#), this equation has an integer solution (b, k) iff $\gcd(a, m) = 1$, that is, a is coprime to m . But we are not done yet - how do we know whether there is a solution b in $\{0, 1, \dots, m-1\}$? Assume $b \geq m$ (similar for $b < 0$). By [A.1.1](#), $b = qm + r$ for some integer q and $0 \leq r < m$. Suppose the equation

$$ab - mk = 1$$

has a solution (b, k) where $b \geq m$. Then

$$ab - mk = a(qm + r) - mk = ar + m(aq - k) = 1 \quad (\star)$$

We claim that $(r, aq - k)$ is also a valid integral solution. That is, there exists $r \in \{0, 1, \dots, m-1\}$ such that $a^{-1} = r$. $\square$

We can now update the definition

$$\mathbb{Z}_m^* = \{a \in \mathbb{Z}_m : \gcd(a, m) = 1\} \quad (1.2.2)$$

Look at the Z_{10} example,

$$Z_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

The subset of elements coprime to 10 is

$$Z_{10}^* = \{1, 3, 7, 9\}$$

We can verify that

$$1^{-1} = 1, \quad 3^{-1} = 7, \quad 7^{-1} = 3, \quad 9^{-1} = 9$$

Exercise 1.2.1

What is Z_p^* if p is prime? (Hint: What is $\gcd(a, p)$ when p is prime and $a < p$?)

Now think about $GL_n(\mathbb{Z}_m)$. We have that $|\mathbb{Z}_m^{n \times n}| = m^{n^2}$, but how many of them are invertible? In this context, a matrix $A \in \mathbb{Z}_m^{n \times n}$ is invertible if there exists $B \in \mathbb{Z}_m^{n \times n}$ such that

$$(AB \bmod m) = I \quad (1.2.3)$$

That is, $\det(A)$ needs to be coprime to m ,

$$GL_n(\mathbb{Z}_m) = \{A \in \mathbb{Z}_m^{n \times n} : \det(A) \in \mathbb{Z}_m^*\} \quad (1.2.4)$$

1.3 Exercises: Group Verification

Exercise 1.3.1 (Groups of Linear Spaces)

Determine whether the following spaces are groups.

1. $(M, \times)$: $M = \{A \in \mathbb{R}^{n \times n} \mid \det(A) \geq 0\}$
2. $(M, \times)$: $M = \{A \in \mathbb{R}^{n \times n} \mid \det(A) = k \text{ for some } k > 0\}$
3. $(M, \times)$ and $(M, +)$: $M = \{A \in \mathbb{R}^{n \times n} \mid \text{trace}(A) = 0\}$
4. $(M, \times)$: $M = \{A \in \mathbb{C}^{n \times n} \mid A \text{ is Hermitian}\}$ (i.e. $A^* = A$)
5. $(M, \times)$: $M = \{A \in \mathbb{C}^{n \times n} \mid A \text{ is Unitary}\}$ (i.e. $A^* = A^{-1}$)
6. $(M, \times)$: $M = \{A \in \mathbb{R}^{n \times n} \mid A \text{ is diagonalizable}\}$
7. $(M, \times)$: $M = \{A \in \mathbb{R}^{n \times n} \mid A \text{ is idempotent}\}$ (i.e. $A^2 = A$)
8. $(M, \times)$: $M = \{A \in \mathbb{R}^{n \times n} \mid A \text{ is idempotent and commutative}\}$ (i.e. $A^2 = A$ and $AB = BA$)
9. $(M, \times)$ and $(M, +)$: $M = \{A \in \mathbb{R}^{n \times n} \mid A \text{ is positive semi-definite (PSD)}\}$ (i.e. $x^T A x \geq 0, \forall x \in \mathbb{R}^n$)
10. $(M, \times)$ and $(M, +)$: $M = \{A \in \mathbb{R}^{n \times n} \mid A \text{ is positive definite (PD)}\}$ (i.e. $x^T A x > 0, \forall x \neq \mathbf{0}_n \in \mathbb{R}^n$)
11. $(V, +)$: $V = \{v \in \mathbb{R}^n \mid Av = \mathbf{0}_m \text{ for some } A \in \mathbb{R}^{m \times n}\}$
12. $(V, +)$: $V = \{v \in \mathbb{R}^n \mid Av \geq \mathbf{0}_n \text{ for some invertible } A \in \mathbb{R}^{n \times n}\}$
13. $(V, +)$: $V = \{v \in \mathbb{R}^n \mid Av \neq \mathbf{0}_m \text{ for some } A \in \mathbb{R}^{m \times n}\}$
14. $(V, \odot)$: $V = \{v \in \mathbb{R}^n \mid v^T x \leq 0\}$ as in (1.3.2), for some $x \geq \mathbf{0}_n$

Exercise 1.3.2 (The Unit Circle)

The unit circle is defined by

$$U = \{z \in \mathbb{C} : |z| = 1\} = \{e^{i\theta} : 0 \leq \theta \leq 2\pi\} \quad (1.3.1)$$

Prove that $(U, \times)$ is an abelian group.

Exercise 1.3.3 (Element-wise Vector Product)

Let $G = \{x \in \mathbb{R}^n : 0 \notin x\}$. We define the operation to be

$$\forall x, y \in \mathbb{R}^n, \quad x \odot y = D(x)y \quad (1.3.2)$$

where $D(x)$ is a $n \times n$ diagonal matrix whose main diagonal is x . Prove that $(G, \odot)$ is an abelian group.

(Hint: The identity element is $e = (1, \dots, 1)^T$.)

Exercise 1.3.4 (Direct Product)

Let $(G, \circ), (H, \star)$ be two groups. We define the direct product to be

$$F = G \times H = \{(g, h) : g \in G, h \in H\} \quad (1.3.3)$$

with operation

$$(g, h) * (g', h') = (g \circ g', h \star h'), \quad \forall (g, h), (g', h') \in F \quad (1.3.4)$$

Note that $|F| = |G| |H|$. Prove that $(F, *)$ is a group.

Exercise 1.3.5 (XOR Group)

The XOR logic applied on two truth values $x, y \in \{T, F\}$ is defined by

$$x \oplus y = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases} \quad (1.3.5)$$

Show that $(\{T, F\}, \oplus)$ is a group. Now we extend this group to binary strings. Let $\mathcal{B}_n$ be the set of all binary strings of length $n \in \mathbb{N}^+$ ($|\mathcal{B}_n| = 2^n$). For example,

$$\mathcal{B}_3 = \{000, 001, 010, 011, 100, 101, 110, 111\}$$

Let $x, y \in \mathcal{B}_n$ where

$$\begin{aligned} x &= x_1 x_2 \cdots x_n, & x_i &\in \{0, 1\} \\ y &= y_1 y_2 \cdots y_n, & y_i &\in \{0, 1\} \end{aligned}$$

Define $x \oplus y = z = z_1 \cdots z_n$ where

$$z_i = x_i \oplus y_i$$

For example,

$$01100 \oplus 10111 = 11011$$

Show that $(\mathcal{B}_n, \oplus)$ is a group.

Exercise 1.3.6 (Group for Symmetric Difference)

Let S be a set. Let 2^S denote the power set of S (note $|2^S| = 2^{|S|}$). For all $A, B \in 2^S$, we define the operation

$$A * B^a = A \cup B \setminus A \cap B$$

Prove that $(2^S, *)$ is an abelian group by showing:

1. $A * B = B * A$ for all $A, B \in 2^S$.
2. $A * B \in 2^S$ for all $A, B \in 2^S$.
3. $(A * B) * C = A * (B * C)$ for all $A, B, C \in 2^S$. (Draw a Venn diagram and shade the areas respectively. Do you get the same shaded regions?)
4. The identity is $\emptyset$.
5. The inverse of $A \in 2^S$ is itself.

^aThis is called the symmetric difference between A and B . Note $x \in A * B$ iff $(x \in A) \oplus (x \in B)$ where $\oplus$ is XOR.

Exercise 1.3.7

Let S be a convex set. The line segment between any $x, y \in \mathbb{R}$ is defined by

$$[x, y] = \{\lambda x + (1 - \lambda)y : \lambda \in [0, 1]\}$$

We also define the mid-point of x, y by $x \oplus y = \frac{1}{2}x + \frac{1}{2}y$. Prove or disprove $(S, \oplus)$ is a group.

(Hint: A set C is convex if for all $x, y \in C$, the line segment joining them lies in C , i.e. $[x, y] \subseteq C$.)

Exercise 1.3.8

Let C be the set of all convex functions of compatible dimension. Prove or disprove $(C, +)$ is a group.

(Hint: $f : \mathbb{R} \mapsto \mathbb{R}$ is convex if $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all $x, y \in \mathbb{R}$ and $\lambda \in [0, 1]$. You can also say that the epigraph under $k \in \mathbb{R}$, denoted $\{f(x) \leq k : x \in \mathbb{R}\}$, is a convex set.)

1.4 Group Properties

1.4.1 Identity

Lemma 1.4.1 (Uniqueness of Identity)

Each group G has a unique identity element.

Proof. Suppose e, f are identity elements of G , so $eg = ge = g$ and $fg = gf = g$ for all $g \in G$. Take $g = f$ then $ef = e$. Take $g = e$ then $fe = e$. Therefore, $f = ef = fe = e$ so the identity element is unique. $\square$

Exercise 1.4.2

Let $g \in G$. Prove that

$$g^2 = g \implies g = e$$

1.4.2 Inverse

Lemma 1.4.3 (Uniqueness of Inverse)

Every element in a group G has a unique inverse in G .

Proof. Fix $g \in G$. Let h, k be inverses of g . By definition, $h, k \in G$ exist.

$$\begin{aligned} gh &= e && \text{as } h \text{ is an inverse of } g \\ k \cdot gh &= k \cdot e && \text{by uniqueness of multiplication} \\ (kg)h &= k \\ eh &= k && \text{as } k \text{ is an inverse of } g \\ h &= k \end{aligned}$$

$\square$

Lemma 1.4.4

Let G be a group and $h, g, k \in G$. If $hg = hk$ or $gh = kh$ then $g = k$.

Proof. By definition, $h^{-1} \in G$ exists for all $h \in G$, so $h^{-1} \cdot hg = h^{-1} \cdot hk \implies g = k$ and $gh \cdot h^{-1} = kh \cdot h^{-1} \implies g = k$. $\square$

Lemma 1.4.5 (Inverse of a Product of Group elements)

Let $g_1, \dots, g_n \in G$. The inverse of $g_1 \cdots g_n$ is $g_n^{-1} \cdots g_1^{-1}$.

Proof. We know $g_i^{-1} \in G$ exists for all element g_i , and by closure property we have $g_n^{-1} \cdots g_1^{-1} \in G$. Consider

$$\begin{aligned} (g_n^{-1} \cdots g_1^{-1})(g_1 \cdots g_n) &= g_n^{-1} \cdots g_2^{-1} \overbrace{(g_1^{-1} g_1)}^e g_2 \cdots g_n \text{ by associativity} \\ &= g_n^{-1} \cdots \overbrace{(g_2^{-1} g_2)}^e \cdots g_n \\ &\dots \\ &= g_n^{-1} g_n \\ &= e \end{aligned}$$

Similarly, $(g_1 \cdots g_n)(g_n^{-1} \cdots g_1^{-1}) = e$. By 1.4.3, such inverse is unique. $\square$

1.4.3 Isomorphisms

Let $(A, \circ), (B, \star)$ be two groups. We say A, B are isomorphic, denoted $A \cong B$, if we can find a bijection (B.2) $f : A \mapsto B$ such that

$$\forall a, a' \in A, \quad f(a \circ a') = f(a) \star f(a') \quad (1.4.1)$$

Equivalently, we can also find

$$\forall b, b' \in B, \quad f^{-1}(b \star b') = f^{-1}(a) \circ f^{-1}(b')$$

For example, consider the multiplication tables of $A = \{\pm 1\}$ and $B = \mathbb{Z}_2$ in (1.4.1).

$\times$	1	-1
1	1	-1
-1	-1	1

A

+	0	1
0	0	1
1	1	0

B

Table 1.4.1: (A) $\{\pm 1\}$ and (B) $\mathbb{Z}_2$

Let us define

$$f(a) = \begin{cases} 0 & \text{if } a = 1 \\ 1 & \text{if } a = -1 \end{cases}$$

which is a valid bijection that relates elements in A and B with no conflicts in the products.

In fact, $A \cong B$ for all groups A, B with $|A| = |B| = 2$. The proof is quite interesting. Let $A = \{e, g\}$ and $B = \{e', g'\}$. It suffices to show that they have the same multiplication table. For A , $eg = ge = g$ and $e^2 = e$ are given by definition. We show that $g^2 = e$. By group closure property, g^2 is one of e, g . Assume a contraction that $g^2 = g$. However,

$$g^2 = g \implies g^2 \cdot g^{-1} = g \cdot g^{-1} \implies g = e$$

Contradiction. Hence $g^2 = e$ is the only case. The unique multiplication table is If we replace e with e' , g with g' , the

$\cdot$	e	g
e	e	g
g	g	e

Table 1.4.2: Multiplication Table of Order-2 Groups

table is still valid. Hence, the bijection f where $f(e) = e'$ and $f(g) = g'$ satisfies (1.4.1) - $A \cong B$. $\square$

Proposition 1.4.6

Any two groups of order 2 are isomorphic to each other.

Similarly, any group of order 3 has this multiplication table upon some bijective mappings,

$\cdot$	e	g	h
e	e	g	h
g	g	h	e
h	h	e	g

Table 1.4.3: Multiplication Table of Order-3 Groups

Example 1.4.7

Prove that any two groups of order 3 are isomorphic to each other. It is sufficient to prove that they have the same multiplication table (1.4.3) upon some bijections.

(Hint: Start by proving $g^2 = h^2 = e$ and $gh = hg = e$.)

Solution (Shortened)

We only prove the hint.

$gh = hg = e$ We know gh is one of e, g, h . If $gh = g$ then $h = e$ (why?). Contradiction. If $gh = h$ then $g = e$ (why?). Contradiction. Thus $gh = e$ only. Therefore, $h = g^{-1}$ so $hg = g^{-1}g = e$. $\square$

$g^2 = h^2 = e$ We know g^2 is one of e, g, h . If $g^2 = g$ then $g = e$ by 1.4.2. Contradiction. If $g^2 = e$ then $g = g^{-1}$, however, we just showed $h = g^{-1}$, so $h = g$. Contradiction. Hence $g^2 = h$ only. Similarly, $h^2 = g$. $\square$

Exercise 1.4.8

Write a complete proof for 1.4.7.

Exercise 1.4.9

Refer to 1.3.5. Prove that $(\mathbb{Z}_2, +) \cong (\{T, F\}, \oplus)$.

For a group $(G, *)$, let

$$G^n = \underbrace{G \times \cdots \times G}_{n \text{ times}}$$

be the chained direct product of itself. Prove that $(G^n, *^n) \cong (\mathcal{B}^n, \oplus)$.

1.4.4 Order**Definition 1.4.10 (Group Order)**

The order of a group G , denoted $|G|$, is the number of elements in G . $|G| = \infty$ if G is an infinite group.

For example,

$$|\mathbb{Z}_m| = m \tag{1.4.2}$$

and

$$|\mathbb{Z}_{10}^*| = 4$$

since $\mathbb{Z}_{10}^* = \{1, 3, 7, 9\}$.

Definition 1.4.11 (Element Order)

Let $g \in G$. The order of g , denoted $|g|$, is the minimum positive integer k such that $g^k = e$. Equivalently,

$$|g| = \inf \{k \in \mathbb{N}^+ \mid g^k = e\}$$

In $\mathbb{Z}_{10}^*$,

$$|1| = 1, \quad |3| = 4, \quad |7| = 4, \quad |9| = 2$$

An interesting observation is that they all divide $|\mathbb{Z}_{10}^*|$.

Proposition 1.4.12

If G is finite, then $|g|$ is finite for all $g \in G$.

Proof. Suppose $k \in \{1, 2, \dots\}$. By property of group closure, we know the sequence g^k is in G . Since G is finite, g^k repeatedly loops over the elements in G and g_k will ultimately go back to some previous term g^m where $m < k$. To prove it, assume a contradiction that g^k is different from any of previous term for all k . However, since all of

$$g, g^2, \dots, \lim_{k \rightarrow \infty} g^k$$

are in G and are different from each other, G should be infinite, which is a contradiction. Hence $g^k = g^m$ for some $m < k$ and thus $g^k = g^m \implies g^{k-m} = e$. We deduce that $|g| \leq k - m$, which is finite. $\square$

Proposition 1.4.13

For all $g \in G$, $g^n = e$ if and only if n is a multiple of $|g|$.

Proof. Let $|g| = k$. By definition $k \leq n$. By A.1.1, $n = kq + r$ for unique $q, r \in \mathbb{Z}$ such that $0 \leq r < k$. Assume for the sake of contradiction that $r \neq 0$. Then

$$\begin{aligned} g^n = e &\implies g^{kq+r} = e \\ &\implies (g^k)^q g^r = e \\ &\implies e^q g^r = e \\ &\implies g^r = e \end{aligned}$$

Since $0 < r < k$, the order should be r , not k . Contradiction. Therefore $r = 0$, and thus $|g| \mid n$. $\square$

Exercise 1.4.14

Suppose group G is abelian. Given $g, h \in G$ and $|g|, |h| < \infty$, show that $|gh| < \infty$. Is it still true when G is not abelian?

Let us find the order of $a \in \mathbb{Z}_m$. Let $k \in \mathbb{Z}$ and we want

$$a^k = (ka \mod m) = 1$$

Therefore, the order of a is

$$|a| = \inf \{k \in \mathbb{N}^+ : ka \equiv 1 \mod m\} \text{ for } a \in \mathbb{Z}_m \quad (1.4.3)$$

Similarly,

$$|a| = \inf \{k \in \mathbb{N}^+ : a^k \equiv 1 \mod m\} \text{ for } a \in \mathbb{Z}_m^* \quad (1.4.4)$$

where we assume

$$\mathbb{N} = \{0, 1, 2, \dots\} \quad (1.4.5)$$

and

$$\mathbb{N}^+ = \mathbb{N} \setminus \{0\} \quad (1.4.6)$$

2 Symmetric Groups

2.1 Definition

Consult [B.2](#) for more information on bijections. Here we focus on bijections on the same set, i.e. $\sigma : A \mapsto A$. We define S_n as the set of all bijections of size n and define the operation as function composition, namely

$$S_n = \{\sigma : \sigma \text{ is a bijection of size } n\} \quad (2.1.1)$$

and

$$\forall \sigma_1, \sigma_2 \in S_n, \quad \sigma_1 \cdot \sigma_2 = \sigma_1 \circ \sigma_2 \quad (2.1.2)$$

It is clear that S_n is non-abelian. An example of $\sigma \in S_5$ is

$$\begin{array}{ll} x & \mapsto \sigma(x) \\ 1 & \mapsto 4 \\ 2 & \mapsto 3 \\ 3 & \mapsto 5 \\ 4 & \mapsto 1 \\ 5 & \mapsto 2 \end{array}$$

We can see that σ changes the order of $\{1, 2, 3, 4, 5\}$ to $\{4, 3, 5, 1, 2\}$, so σ is also called a permutation. Therefore S_n has $n!$ different permutations, i.e. $|S_n| = n!$.

We observe that for all $\sigma_1, \sigma_2 \in S_n$, $\sigma_1 \circ \sigma_2$ is still bijection (which is [B.2.10](#)). To prove it, let $\sigma_3 = \sigma_1 \circ \sigma_2$. We know σ_2 maps $x \in \{1, \dots, n\}$ to a unique $y \in \{1, \dots, n\}$, and then σ_1 maps y to a unique $z \in \{1, \dots, n\}$, so $\sigma_3(x) = \sigma_1 \circ \sigma_2(x) = \sigma_1(\sigma_2(x)) = \sigma_1(y) = z$, such that $\sigma_1 \circ \sigma_2$ uniquely maps x to $z \in \{1, \dots, n\}$ and similarly $(\sigma_1 \circ \sigma_2)^{-1} = \sigma_2^{-1} \circ \sigma_1^{-1}$ uniquely maps z to x . Therefore $\sigma_3 = \sigma_1 \circ \sigma_2$ is still a bijection, which is closed under composition. Also note that the identity function is $e(x) = x : \{1, \dots, n\} \mapsto \{1, \dots, n\}$. By [B.2.3](#), the inverse of a bijection exists. Along with [B.2.11](#), we can show that S_n is a finite group whose operation is the function composition.

2.2 Cycle Form

An easier way to represent σ is to write it in cycle form. It is more like repeatedly applying function composition $\sigma \circ \dots \circ \sigma$ until we find some cycle. We start with 1.

$$\begin{array}{ll} 1 : & (1 \\ \sigma(1) = 4 : & (14 \\ \sigma(4) = 1 : & (14) \text{ [close brackets since successfully found a cycle } 1 - 4 - 1 - 4 \dots] \\ \text{now pick a number not in } (14) & \\ 2 : & (2 \\ \sigma(2) = 3 : & (23 \\ \sigma(3) = 5 : & (235 \\ \sigma(5) = 2 : & (235) \text{ [close brackets since successfully found a cycle } 2 - 3 - 5 - 2 - 3 - 5 \dots] \end{array}$$

So the cycle form is

$$\sigma = (14)(235)$$

We observe that $(235) = (352) = (523)$. Note in cycle form, $\sigma(x)$ returns the number to the right of x in that cycle. For example, if $\sigma = (235)$ then $\sigma(2) = 3, \sigma(3) = 5, \sigma(5) = 2$. If x is not present in the cycle, just skip it and look at the other cycles. For example, $\sigma = (14)(235)$ then $\sigma(1) = (14)(235)[1] = (14)[1] = 4$. Following this logic, the cycles are commutative if they are disjoint, e.g. $(14)(235) = (235)(14)$.

^{2.1.1}by [1.4.5](#)

Given σ_1, σ_2 , how to determine $\sigma_3 = \sigma_1 \circ \sigma_2$? For example, let $\sigma_1 = (25)(134)$, $\sigma_2 = (142)(3)(5)$. We can simply put them together in the specified order,

$$\sigma_3 = \underbrace{(25)(134)}_{\sigma_1} \underbrace{(142)(3)(5)}_{\sigma_2}$$

This is an example of non-disjoint cycles. Recall that σ_3 is still a bijection, so we wish to find a disjoint cycle-form for it, just like for σ_1, σ_2 . We can draw the diagram

$$\begin{array}{ccccc} x & \mapsto & \sigma_2(x) & \mapsto & \sigma_1(\sigma_2(x)) = \sigma_3(x) \\ 1 & \mapsto & 4 & \mapsto & 1 \\ 2 & \mapsto & 1 & \mapsto & 3 \\ 3 & \mapsto & 3 & \mapsto & 4 \\ 4 & \mapsto & 2 & \mapsto & 5 \\ 5 & \mapsto & 5 & \mapsto & 2 \end{array}$$

If we skip the internal process of σ_2 , then we find the mapping table for σ_3 is

$$\begin{array}{cc} x & \mapsto \sigma_3(x) \\ 1 & \mapsto 1 \\ 2 & \mapsto 3 \\ 3 & \mapsto 4 \\ 4 & \mapsto 5 \\ 5 & \mapsto 2 \end{array}$$

whose cycle form is

$$\sigma_3 = (1)(2345)$$

We can efficiently do this with the cycle form and without the use of mapping table.

Example 2.2.1

We wish to write $(25)(134)(142)(3)(5)$ as a disjoint product of cycles.

$$\sigma_3 = \underbrace{(25)}_E \underbrace{(134)}_D \underbrace{(142)}_C \underbrace{(3)}_B \underbrace{(5)}_A$$

Start with 1. Feed it to the cycles one by one in order of $A \rightarrow E$.

Record (1

A : 1 is not in A . Skip.

B : 1 is not in B . Skip.

C : $1 \rightarrow 4$.

D : $4 \rightarrow 1$.

E : 1 is not in E . Skip.

All cycles traversed. Record (1)

Pick 2 which is not present in (1). Start over with 2.

A : 2 is not in A . Skip.

B : 2 is not in B . Skip.

C : $2 \rightarrow 1$.

D : $1 \rightarrow 3$.

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    E : 3 is not in E. Skip.
    All cycles traversed. Record (1)(23

Start over with 3.
A : Skip.
B : 3 → 3.
C : Skip.
D : 3 → 4.
E : Skip.
All cycles traversed. Record (1)(234

Start over with 4.
A : Skip.
B : Skip.
C : 4 → 2.
D : Skip.
E : 2 → 5.
Record (1)(2345

A : 5 → 5.
B : Skip.
C : Skip.
D : Skip.
E : 5 → 2.
Record (1)(2345)

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This algorithm gives the disjoint cycle-form of σ_3 . One can show that its inverse is

$$\sigma_3^{-1} = (5)^{-1}(3)^{-1}(142)^{-1}(134)^{-1}(25)^{-1} = (5)(3)(124)(143)(52)$$

or equivalently

$$\sigma_3^{-1} = (2345)^{-1}(1)^{-1} = (1)^{-1}(2345)^{-1} = (1)(2543)$$

as disjoint cycles are commutative. In general,

$$(a_1 \cdots a_k)^{-1} = (a_k \cdots a_1) \quad (2.2.1)$$

and if $\sigma = \sigma_1 \cdots \sigma_k$ is a product of k permutations (not necessarily disjoint!), its inverse is

$$\sigma = \sigma_1 \cdots \sigma_k \implies \sigma^{-1} = \sigma_k^{-1} \cdots \sigma_1^{-1} \quad (2.2.2)$$

Also note that the identity in S_n is

$$\sigma_e = (1)(2) \cdots (n) \quad (2.2.3)$$

2.3 k -Cycles

Definition 2.3.1 (k -cycle)

A cycle of k different numbers is called a k -cycle.

For example, (123) is a 3-cycle, (24) is a 2-cycle, and (1) is a 1-cycle. Note that a 1-cycles can be ignored in the cycle-form, e.g. $(235)(1)(4) = (235)$.

Definition 2.3.2 (Transpositions)

A transposition is a 2-cycle.

An interesting fact is that every permutation is a product of some transpositions. One possible factorization for a single cycle is

$$(a_1 \cdots a_k) = (a_1 a_k)(a_1 a_{k-1}) \cdots (a_1 a_2) \quad (2.3.1)$$

We can apply this factorization to each cycle in the product. For example,

$$\sigma = (123)(245) = (13)(12)(25)(24)$$

Equivalently,

$$(123)(245) = (12453) = (13)(15)(14)(12)$$

We observe that the product of transpositions is not necessarily disjoint.

Proposition 2.3.3

Every $\sigma \in S_n$ can be written as a product of transpositions.

Proof. We prove that (2.3.1) is a valid transposition factorization. Let $c = (a_1 \cdots a_k)$ be a single-cycle (Note $c \in S_n$ as well). We claim that $c(a_i) = a_{i+1}$ where $1 \leq i \leq k$ and $a_{k+1} = a_1$. We want to show that the product of 2-cycles defined by $c' = (a_1 a_k)(a_1 a_{k-1}) \cdots (a_1 a_2)$ is identical to c .

First look at some extreme cases. We expect $c'(a_1) = a_2$, $c'(a_2) = a_3$, and $c'(a_k) = a_1$. Note that

$$(a_1 a_k)(a_1 a_{k-1}) \cdots (a_1 a_3)(a_1 a_2)[a_1]^a = \underbrace{(a_1 a_k)(a_1 a_{k-1}) \cdots (a_1 a_3)}_{\text{does not contain } a_2} [a_2] = a_2$$

and

$$(a_1 a_k)(a_1 a_{k-1}) \cdots (a_1 a_3)(a_1 a_2)[a_2] = \underbrace{(a_1 a_k)(a_1 a_{k-1}) \cdots (a_1 a_3)}_{\text{does not contain } a_3} \underbrace{[a_1]}_{=a_3} = a_3$$

and

$$(a_1 a_k) \underbrace{(a_1 a_{k-1}) \cdots (a_1 a_3)(a_1 a_2)}_{\text{does not contain } a_k} [a_k] = (a_1 a_k)[a_k] = a_1$$

It works perfectly on a_1 , a_2 and a_k . Now consider the general a_i where $2 < i < k$ such that

$$(a_1 a_k) \cdots (a_1 a_{i+1})(a_1 a_i) \underbrace{(a_1 a_{i-1}) \cdots (a_1 a_2)}_{\text{does not contain } a_i} [a_i] = \underbrace{(a_1 a_k) \cdots (a_1 a_{i+1})}_{\text{no } a_{i+1}} \underbrace{(a_1 a_i)}_{=a_{i+1}} [a_i] = a_{i+1}$$

which is exactly what we expected, $c'(a_i) = a_{i+1}$. Thus we deduce that any single-cycle can be written as a product of 2-cycles.

Every $\sigma \in S_n$ can be represented as a product of single-cycles; each can be written as a product of 2-cycles. Therefore, σ can finally be written as a product of 2-cycles. $\square$

^aThis notation means we feed a_1 to the product of cycles. It is the same as $c'(a_1)$.

2.4 Order

Suppose

$$c = (1452)$$

is a single 4-cycle in S_5 . We wonder the minimum k such that

$$c^k = \sigma_e = (1)(2)(3)(4)(5)$$

Note that

$$\begin{aligned} (1452)(1452) &= (15)(24) \\ (1452)^3 &= (15)(24)(1452) = (1254) \\ (1452)^4 &= (1254)(1452) = (1)(2)(4)(5) = \sigma_e \end{aligned}$$

as $(1254) = (1452)^{-1}$. So $|(1452)| = 4$. Interestingly, the order is just the length of that cycle. As an example, let us feed 2 to the cycle,

$$(1452)[2] = 1$$

We can see that it maps 2 to its next digit in the cycle, which is 1. Now feed 1 to it,

$$(1452)[1] = 4$$

Similarly, it also maps 1 to the next digit 4. As well,

$$(1452)[4] = 5 \text{ and } (1452)[5] = 2$$

which goes back to 2 after 4 repetitions. Following this logic, if we apply c^m on some digit $d \in c$, it changes d to the m th digit after d in c . That said, if repeat it $|c|$ times, d will shift to its original position, which is the identity. This explains why the order of a k -cycle is its length, k .

Lemma 2.4.1 (Order of k -Cycles)

In S_n , the order of a k -cycle is k .

Proof. Let $c = (a_1 \cdots a_k)$ be a k -cycle, $k \geq 2$. It follows by definition that

$$(a_1 \cdots a_k)[a_i] = a_{i+1}, \text{ for all } 1 \leq i \leq k \text{ and } a_{k+i} = a_i \quad (2.4.1)$$

Recursively,

$$(a_1 \cdots a_k)^m[a_i] = a_{i+m} \text{ for all } m \geq 2 \text{ and } 1 \leq i \leq k \quad (2.4.2)$$

We observe that if take $m = k$,

$$(a_1 \cdots a_k)^k[a_i] = a_{i+k} = a_i, \text{ for all } 1 \leq i \leq k$$

Since $\sigma_e(a_i) = a_i$ for all $1 \leq i \leq k$, we have $(a_1 \cdots a_k)^k = \sigma_e$.

Now we show that k is the smallest option for $(a_1 \cdots a_k)^m = \sigma_e$. Assume a contraction that there exists $1 \leq k' < k$ such that $(a_1 \cdots a_k)^{k'} = \sigma_e$. Then

$$(a_1 \cdots a_k)^{k'}[a_i] = a_i$$

However, by (2.4.2),

$$(a_1 \cdots a_k)^{k'}[a_i] = a_{i+k'}$$

This means

$$a_i = a_{i+k'} \text{ for } k' < k$$

which is a contradiction since $a_i, a_{i+k'}$ are two of the $a_1, \dots, a_k$ and should be distinct digits^b.

Therefore, the order of $(a_1 \cdots a_k)$ is k . □

^athe identity of S_n , as defined in (2.2.3)

^bRecall: In a cycle, a digit d repeats every k digits.

Instead of single-cycles, now consider a product of cycles such as

$$\sigma = (125)(36)(47)$$

By 2.4.1,

$$(125)^3 = (36)^2 = (47)^2 = \sigma_e$$

What might be the order of σ ? Note that

$$\sigma^k = (125)^k (36)^k (47)^k$$

since disjoint cycles are commutative. We want it to be σ_e , so we need a some integer k such that $(125)^k$ is a power of $(125)^3$, $(36)^k$ is a power of $(36)^2$, and $(47)^k$ is a power of $(47)^2$. Hence, we can use the least common multiple

$$\text{lcm}(n_1, \dots, n_m) = \inf \{k \in \mathbb{N} : n_1 \mid k, \dots, n_m \mid k\} \quad (2.4.3)$$

Here, let's take

$$\begin{aligned} k &= \text{lcm} \{|(125)|, |(36)|, |(47)|\} \\ &= \text{lcm} \{3, 2, 2\} \\ &= 6 \end{aligned}$$

So

$$\begin{aligned} \sigma^6 &= (125)^6 (36)^6 (47)^6 \\ &= ((125)^3)^2 ((36)^2)^3 ((47)^2)^3 \\ &= \sigma_e^2 \sigma_e^3 \sigma_e^3 \\ &= \sigma_e \end{aligned}$$

Proposition 2.4.2

Let $\sigma = \sigma_1 \cdots \sigma_k \in S_n$ where $\sigma_1, \dots, \sigma_k$ are disjoint. Then

$$|\sigma| = \text{lcm}(|\sigma_1|, \dots, |\sigma_k|)$$

Proof. Let $r = \text{lcm}(|\sigma_1|, \dots, |\sigma_k|)$. Let $n_i = |\sigma_i|$ for $1 \leq i \leq k$. Then $r = n_i m_i$ for some integer m_i .

$$\begin{aligned} (\sigma_1 \cdots \sigma_k)^r &= \sigma_1^r \cdots \sigma_k^r \\ &= \sigma_1^{n_1 m_1} \cdots \sigma_k^{n_k m_k} \\ &= \sigma_e^{m_1} \cdots \sigma_e^{m_k} \\ &= \sigma_e \end{aligned}$$

Let $\sigma^g = \sigma_e$ for some integer g . By 1.4.13, g is a multiple of $n_1, \dots, n_k$. Since r is their least common multiple, $k \leq g$. Therefore, the order of σ is $r = \text{lcm}(|\sigma_1|, \dots, |\sigma_k|)$. □

Exercise 2.4.3

Answer the following questions.

1. Find their orders.
 - (a) $(145)(67)$
 - (b) $(26)(13)(745)$
 - (c) $(142)(423)(39)$
2.
 - (a) Let $\sigma_1 = (136)(25), \sigma_2 = (47)$. Find $|\sigma_1\sigma_2|$. Is it $\text{lcm}(|\sigma_1|, |\sigma_2|)$?
 - (b) Let $\sigma_1 = (136), \sigma_2 = (25)(47)$. Find $|\sigma_1\sigma_2|$. Does it give the same result as in the previous question?

2.5 Fun: Counting k -cycles

In S_n , how many k -cycles are there? Start with some small examples,

$$S_3 = \{e, (12), (13), (23), (123), (132)\}$$

So S_3 has three 2-cycles and two 3-cycles. In general, a k -cycle takes k distinct integers from $\{1, \dots, n\}$, so there are $\binom{n}{k}$ possible combinations of numbers. Now we count the possible permutations of each combination. Note that

$$(a_1 a_2 a_3 a_4) = (a_2 a_1 a_3 a_4) = (a_2 a_3 a_1 a_4) = (a_2 a_3 a_4 a_1)$$

are equivalent. We want to find out what common feature makes them equivalent. Observe that if we exclude a_1 from the strings, the rest digits have the same permutation $(a_2 a_3 a_4)$. Otherwise they are distinct, for example,

$$(a_1 a_2 a_3) \neq (a_1 a_3 a_2) \quad (a_2 a_1 a_3) \neq (a_3 a_1 a_2) \quad (a_2 a_3 a_1) \neq (a_3 a_2 a_1)$$

Therefore, we only need to count the permutations of the digits except for a_1 . Formally speaking, for each combination of k digits, we have $k!$ possible permutations like

$$(a_1 a_2 \cdots a_k)$$

and we then need to exclude those duplicated ones. We can separate all $k!$ permutations into k types by adjusting the position of a_1 , namely

$$\underbrace{(a_1 \square \cdots \square)}_{\text{group 1}} \quad \underbrace{(\square a_1 \square \cdots \square)}_{\text{group 2}} \quad \cdots \quad \underbrace{(\square \cdots \square a_1)}_{\text{group } k}$$

A $\square$ represents a digit in $\{a_2, \dots, a_k\}$. We have seen that each different permutation of these $\square$'s yields a different k -cycle, so each group have $(k-1)!$ k -cycles. However, these k groups yield the same set of permutations, so we have $(k-1)!$ different k -cycles in total. Combining with the $\binom{n}{k}$ combinations of digits, we have

$$\# \text{ } k\text{-cycles in } S_n = \binom{n}{k} (k-1)! = \frac{n!}{k(n-k)!} \quad (2.5.1)$$

Remark

Note that (2.5.1) says k digits can make $(k-1)!$ distinct k -cycles.

Now look at a harder example. Since each $\sigma \in S_n$ can be written as a product of disjoint cycles, suppose σ has n_1 1-cycles, n_2 2-cycles, $\dots$, and n_k k -cycles such that

$$n_1 + 2n_2 + 3n_3 + \cdots + kn_k = n$$

We want to count the total number of distinct such σ . We first need to allocate the n integers to the cycles. Note that 1-cycles need n_1 numbers, 2-cycles need $2n_2$ numbers, $\dots$, and k -cycles need kn_k numbers in total, so the total number of allocations is

$$\binom{n!}{n_1, 2n_2, \dots, kn_k} = \frac{n!}{n_1!(2n_2)! \dots (kn_k)!}$$

which follows a multinomial equation. For each $r \in \{1, \dots, k\}$, we need to allocate the rn_r numbers to the n_r r -cycles, having a total of

$$\binom{rn_r!}{\underbrace{r, \dots, r}_{n_r \text{ of them}}} = \frac{(rn_r)!}{r^{n_r}}$$

different distributions. However, there can be duplicates. Suppose we allocate $\{1, 2, 3, 4\}$ to two 2-cycles and there are $\frac{4!}{2^2} = 6$ distributions, namely

$$\begin{aligned} (12)(34) \\ (13)(24) \\ (14)(23) \\ (23)(14) \\ (24)(13) \\ (34)(12) \end{aligned}$$

We find that $(12)(34) = (34)(12)$, $(13)(24) = (24)(13)$, $(14)(23) = (23)(14)$, so there are actually 3 different allocations, not 6. In general, if the part of r -cycles can be written as

$$c_1 c_2 c_3 \dots c_{n_r}$$

where the c 's are disjoint r -cycles, then among the $\frac{(rn_r)!}{r^{n_r}}$ total allocations, it is equivalent to

$$\begin{aligned} c_1 c_3 c_2 \dots c_{n_r} \\ c_2 c_1 c_3 \dots c_{n_r} \\ c_2 c_3 c_1 \dots c_{n_r} \\ \dots \end{aligned}$$

which are the permutations of $c_1, c_2, \dots, c_{n_r}$ and there are thus $n_r!$ duplicates. We eliminate this order by division, hence the part of r -cycles have

$$\frac{(rn_r)!}{r^{n_r}(n_r!)}$$

different possibilities. Therefore, the total number of distinct σ as a product of disjoint cycles is

$$\begin{aligned} \#\sigma \text{ as a disjoint product} &= \frac{n!}{n_1!(2n_2)! \dots (kn_k)!} \cdot \frac{n_1!}{1^{n_1}(n_1!)} \frac{(2n_2)!}{2^{n_2}(n_2)!} \dots \frac{(kn_k)!}{k^{n_k}(n_k)!} \\ &= \frac{n!}{1^{n_1} 2^{n_2} \dots k^{n_k} (n_1)! (n_2)! \dots (n_k)!} \\ &= \frac{n!}{\prod_{r=1}^k r^{n_r} (n_r)!} \end{aligned} \tag{2.5.2}$$

Exercise 2.5.1

Use (2.5.2) to derive (2.5.1).

Exercise 2.5.2

Fix $r \in \{1, \dots, n\}$. Suppose we randomly pick some $\sigma \in S_n$ and write it as a disjoint product of cycles. What is the probability that it only contains cycles of lengths less than or equal to r ?

(Hint: You might want to find all possibilities for $n_1 + 2n_2 + \dots + rn_r \leq n$ and $n_{r+1} = \dots = n_k = 0$.)

3 Dihedral Groups

3.1 Definition

Suppose we are playing with a paper triangle placed on desk. We can do two things:

1. Rotate it by 60° in the counterclockwise direction^{3.1.1}, or
2. Flip it, so the side originally facing up is now facing down.

Each different posture or position upon combination of operations is called a *state* of the triangle. A triangle goes back to the original state if we rotate it three times or flip it twice. If we don't flip the triangle, there are three states, i.e. by rotating it once, twice, plus the original state. If we flip it, there are also 3 different states, so a triangle has $2 \times 3 = 6$ states in total.

The real situation can be more complicated - what is the state if we rotate the triangle twice, then flip it, and rotate once again? We know it must be one of the 6 states. Does this suggest a potential group? Let r be a rotation and s be a reflection. Intuitively the group can be defined by

$$D_3 = \{r, s \mid r^3 = 1, s^2 = 1\}$$

But wait - since it is not commutative, rs might have an infinite order. As $(rs)^k$ for all $k \in \mathbb{N}$ is one of the finite states, this definition is not sufficient and does not necessarily represent the group that we expected. However, we can add some constraints to rs to limit its order. We claim that

$$rsrs = 1$$

That is, if we apply $rsrs$ to any state, it will ultimately be the same state. Let's prove it. We first rotate the triangle counterclockwisely and then flip it. Now any rotation is clockwise (why?) and thus the second rotation reverses the first rotation. Finally, we flip it back which gives the original state. This shows that $|rs| = 1$. Note that we can also write

$$rsrs = 1 \implies r^{-1} \cdot rsrs \cdot s^{-1} = r^{-1}s^{-1} \implies sr = r^{-1}s \text{ or } rs = sr^{-1}$$

The true definition is

$$D_3 = \{r^3 = 1, s^2 = 1, sr = r^{-1}s\} \quad (3.1.1)$$

Now we can go back to the previous question, 'What is the state if we rotate the triangle twice, then flip it, and rotate once again?' It can be written as r^2sr . Using the definition (3.1.1),

$$\begin{aligned} r^2sr &= r^2(sr) \text{ by associativity} \\ &= r^2(r^{-1}s) \\ &= rs \end{aligned}$$

It is the same as rotating and flipping it! Now try a hard example

$$r^9s^{-5}r^7sr^{-8}$$

A trick is simplifying them as much as possible so that the long string is reduced to the minimum number of r, s .

$$\begin{aligned} r^9s^{-5}r^7sr^{-8} &= \cancel{(r^3)^3} \cancel{(s^2)^{-2}} s^{-1} \cancel{(r^3)^2} rs \cancel{(r^3)^2} r^2 \\ &= s^{-1}rsr^2 = (sr)sr^2 \\ &= r^{-1}ssr^2 = r^{-1}s^2r^2 \\ &= r^{-1}r^2 = r \end{aligned}$$

^{3.1.1}You can also do the clockwise direction. Just make sure you always do the same direction.

We can have more fun,

$$r^3 = 1 \implies r(r^2) = 1 \implies r^{-1} = r^2$$

The similar idea can be applied to all dihedral n -gon for $n \geq 3$. Suppose r is a rotation by an angle of $\frac{2\pi}{n}$ and s is a reflection.

$$D_n = \{r, s \mid r^n = 1, s^2 = 1, sr = r^{-1}s\} \quad (3.1.2)$$

Note that $|D_n| = 2n$ and D_n is non-abelian.

Exercise 3.1.1

Let $r, s \in D_n$. Show that $r^{-k} = r^{n-k}$ for $k \in \mathbb{Z}$.

Exercise 3.1.2

Prove that $sr^k = r^{-k}s$ for any integer k .

3.2 Order

Let p be a state in D_n . We say p is a reflection if $p^2 = 1$.

Exercise 3.2.1

Let $p = sr^k$ for any integer k . Prove that p is a reflection.

Exercise 3.2.2

Let $p = r^k \in D_n$ for any integer k . Let $z = (k \bmod n) \in \mathbb{Z}_n$. Prove that $|p| = |z|$.

(Hint: Let $D_n^* = \{1, r, r^2, \dots, r^{n-1}\}$ be the set of all rotations in D_n (not including any reflections!). Start by showing $D_n^* \cong \mathbb{Z}_n$ and find the bijection.)

In D_5 , consider the order of

$$r^4sr^{-3}$$

It will be very inefficient if we find it by brute forcing

$$(r^4sr^{-3})(r^4sr^{-3}) \dots$$

and stop when it equals 1. However, note that it can only be one of the 10 states in D_5 ,

$$D_5 = \{1, r, r^2, r^3, r^4, s, sr, sr^2, sr^3, sr^4\}$$

Why? Note that if we don't flip it, there are 5 states $1, r, r^2, r^3, r^4$ gotten by rotating zero, one, two, three, four times respectively (it goes back to the original state if you rotate five times) or equivalently $1, r^{-1}, r^{-2}, r^{-3}, r^{-4}$ if you do reverse rotations. If we flip it, similarly there are also 5 states s, sr, sr^2, sr^3, sr^4 upon rotations or equivalently $s, sr^{-1}, sr^{-2}, sr^{-3}, sr^{-4}$ upon reverse rotations. Therefore, if we know the order of these 10 states, we can reduce any $p \in D_5$ to one of them and find its order based on that. For example,

$$r^4sr^{-3} = sr^{-2}$$

Aside, I didn't even need to evaluate it. Let white denote the page originally facing up and black denote the page originally facing down. r^4 rotates the white side 4 times and reversely rotates the black side 4 times. The following s makes the upside down, so now the black side is facing up. Now, another r would reversely rotate the white side once and rotate the black side once, so r^{-3} rotates the white side three times and reversely rotates the black side three times. Let's count the rotations applied on the black side, which is now we are on. It was reversely rotated 4 times and again reversely rotated 3 times, so it experienced 7 reverse rotations in total, namely sr^{-7} . Recall from 3.1.1, $sr^{-7} = sr^3$, and by 3.2.1, its order is 2. Done. Also note that if the reduced string is a rotation, we can find its order using 3.2.2.

4 The Lagrange's Theorem

4.1 Subgroups

We showed before that $(\mathbb{R}, +)$ is a group. Not all subsets of it are a group, but for example, the subset of integers $(\mathbb{Z}, +)$ is also a group. As another example, we know $(\{\pm 1, \pm i\}, \times)$ is a group and its subset $(\{\pm 1\}, \times)$ is also a group. This leads to the definition of subgroups.

4.1.1 Definition

Definition 4.1.1 (Subgroups)

Let $(G, *)$ be a group. A subset H of G is called a subgroup of $(G, *)$, denoted $H \leq G$, if $(H, *)$ is also a group.

The subgroup H of G must satisfy all properties in 1.1.1, namely

1. Closure of Operation: $\forall a, b \in H, a * b \in H$.
2. Associativity: $\forall a, b, c \in H, (a * b) * c = a * (b * c)$.
3. Existence of Identity Element: $\exists e \in H$ such that $ea = ae = a$ for all $a \in H$.
4. Invertibility: $\forall a \in H$, there exists $a^{-1} \in H$ such that $a^{-1} * a = a * a^{-1}$.

A straightforward subgroup of G is $\{e\}$. As well, G is a trivial subgroup of itself.

4.1.2 Subgroup Tests

Lemma 4.1.2 (Subgroup Test)

Let $H \subseteq G$. H is a subgroup of G if and only if H satisfies

1. $H \neq \emptyset$
2. $\forall a, b \in H, a * b \in H$
3. $\forall a \in H, a^{-1} \in H$

Proof. If H is a group then it must satisfy these properties (why?). We only need to prove that a subset satisfying these properties is a valid subgroup. First, an empty set cannot be a group - property 1 is fair. Then, all elements of G satisfy associativity, including those in H - thus H is associative. Properties 2,3 are assuring the closure of operation and invertibility, and by combining them you can have $a * a^{-1} = e \in H$. Therefore, H is a qualified group, and thus a subgroup of G . $\square$

Let us look at an interesting example. Recall the unit circle defined on multiplication is a group,

$$U = \{e^{i\theta} : 0 \leq \theta \leq 2\pi\}$$

Now define the subgroup as

$$U_N = \{e^{i(2\pi k/N)} : k = 0, 1, \dots, N-1\}$$

with $N \in \mathbb{N}^+$. This is called the set of N -th root of unity. For example, when $N = 4$,

$$U_4 = \{e^{i(k\pi/2)} : k = 0, 1, 2, 3\} = \{1, -1, i, -i\}$$

We show U_N is a subgroup of U using 4.1.2. For simplicity we define

$$\theta_N(k) = e^{i(2\pi k/N)}, k = 0, 1, \dots, N-1 \quad (4.1.1)$$

where $\theta(N) = \theta(0) = 1, \theta(k) \cdot \theta(h) = \theta(k + h)$. So U_N can be rewritten as

$$U_N = \{\theta_N(k) : k = 0, 1, \dots, N - 1\}$$

Let $\theta(k), \theta(h) \in U_N$ where $k, h \in \{0, \dots, N - 1\}$. Note by A.1.1,

$$k + h = mN + r \text{ for some } m \in \mathbb{N}, 0 \leq r < N \in \mathbb{N}$$

So we have

$$\begin{aligned} \theta(k) \cdot \theta(h) &= e^{i(2\pi(mN+r)/N)} \\ &= e^{i2\pi m(N/N)} e^{i(2\pi r/N)} \\ &= \underbrace{(e^{i2\pi})^m}_{=1} e^{i(2\pi r/N)} \\ &= e^{i(2\pi r/N)} \\ &= \theta(r) \in U_N \end{aligned}$$

So U_N is closed under multiplication and thus satisfies property 2 in 4.1.2. For property 3 which is the invertibility, we define $\theta(k)^{-1} = \theta(g)$ where

$$e^{i(2\pi(k+g)/N)} = 1$$

We know such $g \in \{0, 1, \dots, N - 1\}$ exists (prove it!), which is the one satisfying

$$(k + g \mod N) = 0$$

So property 3 in 4.1.2 is satisfied. Therefore, U_N is a valid subgroup of U .

Corollary 4.1.3

An important property used above is $U_N \cong \mathbb{Z}_N$. A bijection could be

$$\theta(k) \in U_N \leftrightarrow k \in \mathbb{Z}_N$$

We now formally rewrite the above example,

Example 4.1.4

Let

$$U = \{e^{i\theta} : 0 \leq \theta \leq 2\pi\}$$

be the unit circle and

$$U_N = \left\{ e^{i(2\pi k/N)} : k = 0, 1, \dots, N - 1 \right\} \quad (4.1.2)$$

be the set of N -th root of unity where $N \in \mathbb{N}^+$. By 4.1.2, U_N is a subgroup of U .

Exercise 4.1.5

Let $p \in \mathbb{N}^+$. Let

$$U^p = \{e^{i\theta} : 0 \leq \theta \leq 2p\pi\} \quad (4.1.3)$$

be the unit circle defined on the extended standard complex plane $[0, 2p\pi]$ (you can think it as a stack of p standard planes) and

$$U_N^p = \left\{ e^{i(2p\pi k/N)} : k = 0, 1, \dots, N - 1 \right\}$$

be the corresponding set of N -th root of unity. In this case,

$$\theta_N^p(k) = e^{i(2p\pi k/N)}$$

Prove or disprove that U_N^p is a subgroup of U^p .

(Hint: It should be very similar to what we did for 4.1.4.)

As a negative example, is $\{3, 5, 8\}$ a subgroup of $\mathbb{Z}_{10}$? No because a group must contain the identity element. Here, 0 is not present in the subset so we can already deduce that it is not a subgroup. You can also see that it does not satisfy properties of closure and invertibility.

Here is an easier test of subgroups (but with more complicated proof!).

Lemma 4.1.6 (Condensed Subgroup Test)

Let $H \subseteq G$. Then $H \leq G$ if and only if H satisfies

1. $H \neq \emptyset$
2. $\forall a, b \in H, ab^{-1} \in H$.

Proof. Suppose the subset H satisfies these properties. We will only prove H satisfies the closure of operation, existence of identity, and invertibility properties.

Existence of Identity Let $a \in H$ be arbitrary. Then

$$aa^{-1} = e \in H$$

Invertibility Let $a \in H$ be arbitrary. Then since $e, a \in H$,

$$ea^{-1} = a^{-1} \in H$$

Closure of Operation Let $a, b \in H$. We have $ab^{-1} \in H$. Since $b^{-1}, b \in H$, we have $b^{-1}b^{-1} = b^{-2} \in H$. Since $ab^{-1}, b^{-2} \in H$, we have $ab^{-1}(b^{-2})^{-1} = ab \in H$.

(the rest of this proof is left as exercise)

□

Exercise 4.1.7

Let $H, K \leq G$.

1. Is $H \cap K \leq G$?

(Hint: H, K must both contain e so $H \cap K \neq \emptyset$. For all $a, b \in H \cap K$, is $ab \in H \cap K$? Note that a, b are both in H and are both in K ; so where is ab ? What about a^{-1}, b^{-1} ?)

2. Is $H \cup K \leq G$?

(Hint: $2\mathbb{Z}, 3\mathbb{Z} \leq \mathbb{Z}$, but is $2 + 3$ in $2\mathbb{Z} \cup 3\mathbb{Z}$?)

4.2 Cyclic Groups

4.2.1 Definition

We first introduce the generator. For a group G , a generator on some $a \in G$ is defined by

$$\langle a \rangle = \{a^k : k \in \mathbb{Z}\} \text{ for some } a \in G \quad (4.2.1)$$

This leads to the definition of cyclic groups.

Definition 4.2.1 (Cyclic Groups)

A group G is cyclic if it can be written as $G = \langle a \rangle$ for some $a \in G$.

Consider $\mathbb{Z}_{10}^* = \{1, 3, 7, 9\}$. Note that

$$\begin{aligned} \langle 3 \rangle &= \{\dots, 3^{-2}, 3^{-1}, 1, 3, 3^2, 3^3, \dots\} \\ &= \{\dots, 9, 7, 1, 3, 9, 7, \dots\} \end{aligned}$$

If we write it out more completely, it is actually a sequence

$$\dots \underbrace{3-9-7-1}_{\text{cycle 1}} - \underbrace{3-9-7-1}_{\text{cycle 2}} - \underbrace{3-9-7-1}_{\text{cycle 3}} \dots$$

Therefore

$$\langle 3 \rangle = \{3, 9, 7, 1\} = \mathbb{Z}_{10}^*$$

As we can see, $\langle 3 \rangle$ repeatedly generates cycles of 3, 9, 7, 1; this is why it's called cyclic. $\langle 7 \rangle$ generate exactly the same sequence,

$$\langle 7 \rangle = \dots \underbrace{7-9-3-1}_{\text{cycle 1}} - \underbrace{7-9-3-1}_{\text{cycle 2}} - \underbrace{7-9-3-1}_{\text{cycle 3}} \dots$$

Thus

$$\mathbb{Z}_{10}^* = \langle 3 \rangle = \langle 7 \rangle$$

However, $\langle 1 \rangle = \{1\}$ and $\langle 9 \rangle = \{1, 9\}$ are not generators of $\mathbb{Z}_{10}^*$. Instead, they are subgroups of G .

Exercise 4.2.2 (Subgroup Generators)

Suppose G is a group. Prove that for all $a \in G$, $\langle a \rangle$ is a subgroup of G , i.e. $\langle a \rangle \leq G$.

(Hint: First show $\langle a \rangle \neq \emptyset$ and $\langle a \rangle \subseteq G$. Then prove $\langle a \rangle$ is a group.)

Now we wonder $|\langle a \rangle|$. Assume $|a| = n < \infty$. Note that

$$\begin{aligned} \langle a \rangle &= \{\dots e, a, a^2, \dots, a^{n-1}, a^n, a^{n+1}, a^{n+2} \dots\} \\ &= \{\dots e, a, a^2, \dots, a^{n-1}, a^n, a^n a, a^n a^2 \dots\} \\ &= \{\dots e, a, a^2, \dots, a^{n-1}, e, a, a^2 \dots\} \text{ since } a^n = e \\ &= \{e, a, a^2, \dots, a^{n-1}\} \end{aligned}$$

Therefore, $|\langle a \rangle| = |a| = n$ and $\langle a \rangle$ contains exactly $e, a, a^2, \dots, a^{n-1}$.

Exercise 4.2.3

Suppose a group G is finite. Prove that for all $a \in G$, $|\langle a \rangle| = |a|$. You might show $\langle a \rangle = \{e, a, a^2, \dots, a^{|a|-1}\}$.

(Hint: By A.1.1, you can decompose all $a^k \in G$ into $a^{m|a|+r} = (a^{|a|})^m a^r = a^r$ for some $0 \leq r < |a|$. For all

$k \in \mathbb{Z}$, how many possible r can we have? You can also show $\langle a \rangle \cong \mathbb{Z}_{|a|}$ and thus $|\langle a \rangle| = |\mathbb{Z}_{|a|}|$.

Exercise 4.2.4 (Cyclic Groups are Abelian)

Show that all cyclic groups are abelian. (Note that non-abelian groups cannot be cyclic!)

Exercise 4.2.5

Let G be cyclic. Show that $G \cong \mathbb{Z}$ if $|G| = \infty$; $G \cong \mathbb{Z}_n$ if $|G| = n < \infty$.

Exercise 4.2.6

Explain why the following groups are not cyclic:

1. D_n and S_n
2. $GL_n(\mathbb{Z})$
3. $\mathbb{Z}_2 \times \mathbb{Z}_2$

Proposition 4.2.7

Let $a \in G$. If $|a| = |G| < \infty$ then $G = \langle a \rangle$, i.e. G is cyclic.

Proof. This is clear if $G = \{e\}$ and $a = e$. Otherwise, let $|G| = n$, and we have $\langle a \rangle = \{e, a, a^2, \dots, a^{n-1}\}$. By 4.2.2, $\langle a \rangle \subseteq G$. Since $|G| = |\langle a \rangle|$ by 4.2.3, G cannot contain elements other than those in $\{e, a, a^2, \dots, a^{n-1}\}$ (otherwise $|G| > n$, which is a contradiction). Therefore, $G = \langle a \rangle$. $\square$

4.2.2 Cyclic Subgroups

Now go back to the previous question that $\langle 9 \rangle$ is a subgroup of $\mathbb{Z}_{10}^*$. Recall

$$\mathbb{Z}_{10}^* = \cdots \underbrace{7-9-3-1}_{\text{cycle 1}} - \underbrace{7-9-3-1}_{\text{cycle 2}} - \underbrace{7-9-3-1}_{\text{cycle 3}} \cdots$$

and

$$\langle 9 \rangle = \cdots \underbrace{9-1}_{\text{cycle 1}} - \underbrace{9-1}_{\text{cycle 2}} - \underbrace{9-1}_{\text{cycle 3}} \cdots$$

We see that $\langle 9 \rangle$ is a sub-cycle of $\mathbb{Z}_{10}^*$ and $|\langle 9 \rangle| = 2$ divides $|\mathbb{Z}_{10}^*| = 4$. We observe that a subgroup of a cyclic group is also cyclic. As a more concrete example, consider the additive integer group

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\} = \langle 1 \rangle = \langle -1 \rangle$$

and its k -subgroup for $k \in \mathbb{Z}$

$$k\mathbb{Z} = \{\dots, -2k, -k, 0, k, 2k, \dots\} = \langle \pm k \rangle \quad (4.2.2)$$

Note that $\langle 0 \rangle = \{0\}$ is trivially a cyclic subgroup and

$$2\mathbb{Z} = \{\dots, -4, -2, 0, 2, 4, \dots\} = \langle 2 \rangle = \langle -2 \rangle$$

is a subgroup of $\mathbb{Z}$ such that $\mathbb{Z} \cong 2\mathbb{Z}$. Clearly, $2\mathbb{Z}$ is also cyclic. The trick is the generator kernel 2. We know that $2\mathbb{Z}$ contains all multiples of 2, which is itself a number generated by $\langle 1 \rangle$. By repeatedly multiplying 2 we can get

the cyclic subgroup $2\mathbb{Z}$. Specifically, any two elements in $2\mathbb{Z}$ can be written as $2k, 2h$ for some $k, h \in \mathbb{Z}$. Clearly, $2k + 2h = 2(k + h) \in 2\mathbb{Z}$ and $2k + (-2k) = 2h + (-2h) = 0$. However, 4 is not a good choice here since $\langle 4 \rangle = \{\dots, -8, -4, 0, 4, 8, \dots\}$. It seems that we can pick the generator kernel as small as possible.

Now look at the general case for any subgroup $H \leq G = \langle a \rangle$. If $H = \{e\}$ then clearly it is cyclic. Assume $H \neq \{e\}$. For all $g \in H$, $g = a^k$ for some $k \in \mathbb{Z}$ since $H \subseteq \langle a \rangle = G$. We want to see whether H is cyclic, i.e.

$$H = \langle a^k \rangle = \{a^{kn} \mid n \in \mathbb{Z}\} \text{ for some } k \in \mathbb{Z}$$

By property of a subgroup, for all $m, n \in \mathbb{Z}$ if $g^m, g^n \in H$ then $g^{m+n} = g^p \in H$ for some $p \in \mathbb{Z}$. The exponent part seems isomorphic to a subgroup of integer group $k\mathbb{Z}$ for some $k \in \mathbb{Z}$, and indeed the bijection is

$$g^m \in H \leftrightarrow m \in k\mathbb{Z} \quad (4.2.3)$$

You can verify this by checking (1.4.1). By (4.2.2), $k\mathbb{Z} = \langle k \rangle$. Why? Note that $k\mathbb{Z}$ can be written as the cyclic form $\langle m \rangle$ for some $m \in k\mathbb{Z}$ if and only if

$$m \mid x \text{ for all } \forall x \in k\mathbb{Z}, \text{ i.e. } m \mid 0, m \mid k, m \mid 2k, \dots \quad (4.2.4)$$

Such m exists and must be k which is the minimum positive integer in $k\mathbb{Z}$ ^{4.2.1}. This suggests $k\mathbb{Z} = \langle k \rangle$. On the other hand, $\langle a^k \rangle \subseteq \langle a^m \rangle$ if $k \mid m$. Anyway, using the isomorphism (4.2.3), we can similarly show that any subgroup H of a cyclic group $G = \langle a \rangle$ where $H \neq \{e\}$ can be written as

$$H = \langle a^k \rangle, \quad k = \inf \{k \in \mathbb{N}^+ \mid a^k \in H\} \quad (4.2.5)$$

We can apply the similar idea to generalized groups.

Theorem 4.2.8

Let $G = \langle a \rangle$ be a cyclic group. Then any $H \leq G$ is cyclic and can be written as

$$H = \langle a^k \rangle, \quad k = \inf \{k \in \mathbb{N}^+ \mid a^k \in H\}$$

Proof. If $H = \{e\}$ then it is trivially cyclic. Now assume $H \neq \{e\}$.

$H \subseteq \langle a^k \rangle$ Let $a^m \in H$ be arbitrary. We want to show $a^m \in \langle a^k \rangle = \{a^{kn} \mid n \in \mathbb{Z}\}$ ^a. By definition, $|k| \leq |m|$ and we want to show $k \mid m$. By A.1.1, $m = kn + r$ for unique $n \in \mathbb{Z}, 0 \leq r < k \in \mathbb{Z}$. Assume a contradiction $r \neq 0$ so now $0 < r < k$. Hence $g^m = g^{kn+r}$ and

$$g^r = \underbrace{g^{-kn}}_{\in H} \underbrace{g^m}_{\in H} \in H$$

But now the smallest positive integer p for $a^p \in H$ should be r , not k . Contradiction. Therefore, $r = 0$ and thus $k \mid m$. It suffices to deduce $a^m \in \langle a^k \rangle$ and thus $H \subseteq \langle a^k \rangle$.

$\langle a^k \rangle \subseteq H$ Let $a^{kn} \in \langle a^k \rangle$ for arbitrary $n \in \mathbb{Z}$. By property of subgroup H , if $a^k \in H$ then $a^{kn} \in H$. Therefore, it follows that $\langle a^k \rangle \subseteq H$. □

^aFrom the above example of $\mathbb{Z}$, we know a^k can "produce" any element in H , like stacking bricks. Here we use the more formal proof by contradiction.

Let $G = \langle a \rangle$ be a cyclic group of order 6,

$$G = \{e, a, a^2, a^3, a^4, a^5\}$$

^{4.2.1}can also be the maximum negative integer $-k$ since $m \mid |k|$

We can show that the distinct subgroups are

$$\begin{aligned}\langle e \rangle &= \langle a^6 \rangle = \{e\} \\ \langle a^3 \rangle &= \{e, a^3\} \\ \langle a^2 \rangle &= \langle a^4 \rangle = \{e, a^2, a^4\} \\ \langle a \rangle &= \langle a^5 \rangle = \{e, a, a^2, \dots, a^{n-1}\} = G\end{aligned}$$

It seems that $\langle a \rangle = \langle a^5 \rangle$ because 5 is coprime to $|G| = 6$, and $\langle a^2 \rangle = \langle a^4 \rangle$ because 2, 4's gcd to 6 are both 2. Is that true for all subgroups?

Proposition 4.2.9

Let $G = \langle a \rangle$ be a finite cyclic group of order n . Then $G = \langle a^k \rangle$ if and only if $k \in \mathbb{Z}$ is coprime to $|G|$, i.e. $\gcd(k, n) = 1$.

Proof. Note that $|\langle a \rangle| = |a| = n$ by 4.2.3. We want to show $\langle a \rangle = \langle a^k \rangle$ if and only if $\gcd(k, n) = 1$. For convenience, we say $a^{pk} \in \langle a^k \rangle$ for $p \in \mathbb{Z}$ instead of $a^p \in \langle a^k \rangle$ where p is a multiple of k .

$\gcd(k, n) = 1 \implies \langle a \rangle = \langle a^k \rangle$ Assume $\gcd(k, n) = 1$. Since $\langle a^k \rangle \subseteq \langle a \rangle$ (why? check definition (4.2.1)), it suffices to show $\langle a \rangle \subseteq \langle a^k \rangle$, that is, $\langle a^k \rangle$ contains every a^z in $\langle a \rangle$. We thus want to show, for all $z \in \mathbb{Z}$, there exists some $p \in \mathbb{Z}$ such that $a^{kp} = a^{(kp \bmod n)} = a^z$, i.e.

$$\forall z \in \mathbb{Z}, \exists p \in \mathbb{Z}, \quad (kp \bmod n) = z$$

if we only look at the components which form a group $\mathbb{Z}_n$ (verify by (1.4.1)). It is equivalent to look at

$$kp - nm = z \tag{*}$$

Note that by A.3.2, a pair of integer solutions (p', m) exists for every $z \in \{1, \dots, n-1\}$ since $\gcd(k, n) = 1$ divides every such z . Therefore, for all $a^z \in \langle a \rangle$, we must have $a^z \in \langle a^k \rangle$ since $a^z = a^{p'k}$ for some integer p' .

$\langle a \rangle = \langle a^k \rangle \implies \gcd(k, n) = 1$ Exercise. (Hint: You might want to show equation (*) should have a solution for $z = 1$, which requires $\gcd(k, n) = 1$ by A.3.2.) $\square$

Remark

To visualize $\langle a \rangle$ and $\langle a^k \rangle$, note that

$$\langle a \rangle = \{e, a, a^2, \dots, a^{n-1}\}$$

and

$$\langle a^k \rangle = \{e, a^k, a^{2k}, \dots, a^{(n-1)k}\} \text{ (duplicates exist!)}$$

Corollary 4.2.10 (Counting Generators of a Cyclic Group)

Let G be a finite cyclic group of order n . Then the number of generators $\langle a^k \rangle$ for some $k \in \{0, \dots, n-1\}$ is $\varphi(n)$, where

$$\begin{aligned}\forall n \in \mathbb{N}^+, \quad \varphi(n) &= |\mathbb{Z}_n^*| = \text{number of integers within } [0, n-1] \text{ that are coprime to } n \\ &= n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right)\end{aligned}$$

if the prime factorization of n is structured as

$$n = p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$$

For prime p , $\varphi(p) = p - 1$.

Example 4.2.11

For example, $25 = 5^2$ so

$$\varphi(25) = 25 \left(1 - \frac{1}{5}\right) = 20$$

and $60 = 2^2 \cdot 3 \cdot 5$ so

$$\varphi(60) = 60 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) = 16$$

Proposition 4.2.12

Let $G = \langle a \rangle$ be finite and $|G| = n$. Then $\langle a^k \rangle = \langle a^{\gcd(k, n)} \rangle$.

Proof. $\langle a^{\gcd(k, n)} \rangle \subseteq \langle a^k \rangle$: Let $p = k / \gcd(k, n)$. Since $k = p \cdot \gcd(k, n)$, by definition (4.2.1), it suffices to show $\langle a^{\gcd(k, n)} \rangle \subseteq \langle a^k \rangle$.

$\langle a^k \rangle \subseteq \langle a^{\gcd(k, n)} \rangle$: We wish to prove that $\langle a^{\gcd(k, n)} \rangle$ can generate any $a^k, k \in \mathbb{Z}$, i.e.

$$\exists x \in \mathbb{Z}, \quad x \cdot \gcd(k, n) \pmod n = k, \forall k \in \mathbb{Z} \quad (\star)$$

We look at the equivalent equation

$$x \gcd(k, n) - yn = k$$

For simplicity, let $t = n / \gcd(k, n), m = \gcd(k, n)$, then

$$xm - y(tm) = k$$

Since $\gcd(m, tm) = m$ which divides k for every $k \in \mathbb{Z}$ (recall $\gcd(k, n) \mid k$), by A.1.1, the equation has a pair of integer solutions (x, y) for each k . We have thus proved $(\star)$. Therefore, $\langle a^k \rangle \subseteq \langle a^{\gcd(k, n)} \rangle$. $\square$

Remark

We can easily prove 4.2.9 using 4.2.12. Try to show it.

Corollary 4.2.13

By 4.2.12, $\langle a^k \rangle = \langle a^l \rangle$ if and only if $\gcd(k, n) = \gcd(l, n)$.

Exercise 4.2.14 (Counting Distinct subgroups of a Cyclic Group)

Let G be a cyclic group of order 12. How many distinct subgroups does it have? Show that it is the number of positive divisors of group order.

(Hint: They are $\langle e \rangle, \langle a \rangle, \langle a^2 \rangle, \langle a^3 \rangle, \langle a^4 \rangle, \langle a^6 \rangle$. Note that $12 = 2^2 \cdot 3$, so there are $(2 + 1)(1 + 1) = 6$ of them.)

We are interested in the size of $\langle a^k \rangle$. Consider

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\} = \langle 1 \rangle$$

Look at $\langle 2 \rangle$ in the cyclic form

$$\begin{aligned}\langle 2 \rangle &= \cdots 2 - 4 - 6 - 8 - 10 - 12 - 14 - 16 \cdots \\ &= \cdots 2 - 4 - 0 - 2 - 4 - 0 - 2 - 4 \cdots \\ &= \cdots 2 - 4 - 0 \cdots\end{aligned}$$

Clearly, the period of the cycle is 3, because if we repeatedly add 2, the sequence follows $2^0 = 0, 2^1 = 2, 2^2 = 4, 2^3 = 0$ which goes back to the identity and thus repeats the cycle after $6/2 = 3$ times. Similarly,

$$\langle 3 \rangle = \cdots 3 - 0 \cdots$$

$$\langle 5 \rangle = \langle 1 \rangle = \cdots 5 - 4 - 3 - 2 - 1 - 0 \cdots$$

Note that we can reduce $\langle k \rangle$ to $\langle \gcd(k, n) \rangle$ by 4.2.12. It seems that $\langle k \rangle$ has $\frac{n}{\gcd(k, n)}$ elements.

Proposition 4.2.15

If $G = \langle a \rangle$ has order n then $|\langle a^k \rangle| = \frac{n}{\gcd(k, n)}$ for all $k \in \mathbb{Z}$.

Proof. Let $m = \gcd(k, n)$. By 4.2.12, we can reduce $\langle a^k \rangle$ to $\langle a^m \rangle$. Let $t = \frac{n}{m}$ which is a positive integer. We claim that

$$\langle a^m \rangle = \{e, a^m, a^{2m}, \dots, a^{(t-1)m}\} \quad (\star)$$

which has order t .

(The rest of this proof is left as an exercise. You can mimic what we did in 4.2.3 to get $(\star)$; that is, for each a^{pm} in $\langle a^m \rangle$ for $p \in \mathbb{Z}$, decompose $pm = (xt + r)m$ for some integer $x, 0 \leq r \leq t - 1$. It can then be reduced to $(\star)$.) $\square$

Exercise 4.2.16

Let $G = \langle a \rangle$ with order n . How many distinct subgroups have order p , where $p \mid n$? The answer is also the number of distinct elements of order p since $|\langle a^k \rangle| = \frac{n}{\gcd(k, n)}$. (Answer: $\varphi(p)$.)

(Hint: By 4.2.15, you might count how many of $k \in \{0, 1, \dots, n-1\}$ satisfy $\frac{n}{\gcd(k, n)} = p$, i.e. $\gcd(k, n) = n/p$. Let $m = n/p$. By A.2.3, $\gcd(k/m, p) = 1$. There are $\varphi(p)$ such k . For example, if $|G| = 60$ and we want to find the number of elements of order 15, we can count those $k \in \{0, 1, \dots, n-1\}$ with $\gcd(k, 60) = 60/15 = 4$, that is, $\gcd(k/4, 15) = 1$. We thus want $4 \mid k$ but $3, 5, 15 \nmid k$. There are $\varphi(15) = 8$ of them, namely a^k where $k = 4, 8, 16, 28, 32, 36, 44, 52, 56$. The formal proof will be decomposed into the following theories.)

^aYou might find that no subgroups has order p if $p \nmid n$. This is the Lagrange's theorem.

At this stage we can look into the baby version of Lagrange's theorem, which is a pivotal property in group theory.

Theorem 4.2.17 (Lagrange's Theorem for Cyclic Groups)

If G is finite and cyclic and if $H \leq G$, then $|H|$ divides $|G|$ and $[G : H] = \gcd(k, n)$.

Proof. Let $|G| = n$. By 4.2.8, if $G = \langle a \rangle$ then $H = \langle a^k \rangle$ for some $k \in \mathbb{N}^+$. By 4.2.15,

$$|H| = |\langle a^k \rangle| = \frac{n}{\gcd(k, n)}$$

which divides $|G| = n$. By division we can get $[G : H] = |G| / |H| = \gcd(k, n)$. $\square$

Corollary 4.2.18 (Lagrange's Theorem for Element Order (Cyclic Groups))

By 4.2.3 and 4.2.17, if G is finite and cyclic then $|a|$ divides $|G|$ for any $a \in G$.

Exercise 4.2.19 (Lagrange's Theorem for Abelian Groups)

Suppose G is an abelian group of order n . Fix $g \in G$. Show that

1. $(gg_1)(gg_2) \cdots (gg_n) = g^n(g_1g_2 \cdots g_n)$
2. $(gg_1)(gg_2) \cdots (gg_n) = g_1g_2 \cdots g_n$ (Hint: By B.2.6, gg_i maps g_i to some unique g'_i in G .)
3. Deduce that $|g| \mid n$. (Hint: You can get $g^n = e$. What does 1.4.13 tell?)

4.2.3 Double Generators

The double generator $\langle a, b \rangle$ for some $a, b \in G$ is the set of all possible permutation products of powers of a, b , i.e.

$$\langle a, b \rangle = \{a^k b^l a^h a^m \cdots \mid k, l, h, m, \dots \in \mathbb{Z}\} = \{\dots, e, a, b, ab, ab^2, a^2b, a^2b^2, \dots\}^{4.2.2} \quad (4.2.6)$$

For example, in D_3 ,

$$\langle r, s \rangle = \{e, r, r^2, s, sr, sr^2\}$$

so $D_3 = \langle r, s \rangle$ with constraints $r^3 = s^2 = 1, rs = sr^{-1}$.

Remark (Aside)

We use a^* to represent all powers of a , i.e.

$$a^* = a^m, \quad \forall m \in \mathbb{Z} \quad (4.2.7)$$

Note that the asterisk $*$ can be filled with integers. So $a^* b^* a^* b^* \cdots$ corresponds to the set $\langle r, s \rangle$. Such $*$ -generator is said to be unambiguous if, for each possible $a^k b^l a^h a^m \cdots$ it can generate, there is only one assignment of integers to each $*$. For example, $a^* b^*$ can generate $a^k b^l$ for all $k, l \in \mathbb{Z}$ and is unambiguous since the only assignment is $(*_1, *_2) = (k, l)$. However, $a^* a^* b^*$ is the same as $a^* b^*$ but is ambiguous since for each $a^k b^l$, we can let $*_1 + *_2 = k, *_3 = l$, which has many possible assignments. We claim that

$$\langle a, b \rangle = a^*(b^* b a^* a)^* b^* \quad (4.2.8)$$

is an unambiguous representation.

Exercise 4.2.20

Show that if G is abelian then

$$\langle a, b \rangle = \{a^k b^l a^h a^m \cdots \mid k, l, h, m, \dots \in \mathbb{Z}\} = \{a^m b^n \mid m, n \in \mathbb{Z}\}$$

If $G = \langle a \rangle$ is cyclic then

$$\langle a^k, a^l \rangle = \{a^{km+ln} \mid m, n \in \mathbb{Z}\}$$

Also show that it's cyclic, i.e.

$$\langle a^k, a^l \rangle = \langle a^p \rangle$$

where $p = \gcd(k, l)$.

^{4.2.2}This is a subgroup of G , just like 4.2.2.

(Hint for last statement: We want

$$\{a^{km+ln} \mid m, n \in \mathbb{Z}\} = \{a^{pt} \mid t \in \mathbb{Z}\}$$

$\subseteq$ is clear since for all $m, n \in \mathbb{Z}$, the linear diophantine equation

$$km + ln = pt$$

is always consistent for some $t \in \mathbb{Z}$ by A.3.2 as $\gcd(k, l) \mid pt$. $\supseteq$ can be shown similarly.)

4.3 Cosets

4.3.1 Definition

Definition 4.3.1 (Cosets)

For a group G , the left coset of subgroup $H \leq G$ defined on $a \in G$ is

$$aH = \{ah \mid h \in G\}^a$$

Similarly, the right coset of H defined on a is

$$Ha = \{ha \mid h \in G\}^b$$

Remark

For convenience, I personally call a the coefficient of coset aH or Ha .

^acan also be written as $a + H = \{a + h \mid h \in G\}$ in additive notation

^bcan also be written as $H + a = \{h + a \mid h \in G\}$ in additive notation

Note that $aH = Ha$ if H is abelian. Also note that H is both a left and a right coset of itself since

$$H = hH = Hh, \quad \forall h \in H \quad (4.3.1)$$

By symmetry of left and right cosets, we can just focus on left cosets.

Remark

If unspecified, a "coset" refers to a "left coset". You can also do right cosets by default but they are basically the same logic.

As an example, consider

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$$

Let

$$H = \{0, 2, 4\}$$

be the subgroup. All possible cosets are

$$\begin{aligned} H = 0 + H = 2 + H = 4 + H &= \{0, 2, 4\} \quad (aH, \forall a \in H) \\ 1 + H = 3 + H = 5 + H &= \{1, 3, 5\} \end{aligned}$$

Note that $1 + H, 3 + H, 5 + H$ are not subgroups; they don't even contain e . This shows that cosets are not necessarily subgroups.

Exercise 4.3.2

We can see that

$$k\mathbb{Z} = \{\dots, -2k, -k, 0, k, 2k, \dots\}$$

for some $k \in \mathbb{N}^+$ is a subgroup of $\mathbb{Z}$. Show that $k\mathbb{Z}$ has k distinct cosets, namely

$$k\mathbb{Z}, \quad 1 + k\mathbb{Z}, \quad 2 + k\mathbb{Z}, \quad \dots, \quad (k-1) + k\mathbb{Z}$$

For another example,

$$\mathbb{Z}_{10}^* = \{1, 3, 7, 9\}$$

and let

$$H = \{1, 9\}$$

be the subgroup. The cosets are

$$H = 1H = 9H = \{1, 9\}$$

$$3H = 7H = \{3, 7\}$$

Similarly, $3H, 7H$ are not subgroups since $3, 7$ are not in H .

4.3.2 Properties of Cosets**Proposition 4.3.3**

Let G be any group and $H \leq G$ be any subgroup of G . Then $H = aH$ iff $a \in H$.

Proof. ($\implies$) Let $H = aH$. Therefore $aH \subseteq H$ and thus $ah \in H$ for all $h \in H$. Let $ah = h'$ for some $h' \in H$, then $a = h'h^{-1} \in H$ by closure and invertibility properties of subgroup H .

($\impliedby$) Assume $a \in H$. Using the similar idea in B.2.4 and B.2.6, ah for $h \in H$ is a bijection to some unique h' in H , so $H = aH$ (that is, aH just reorders elements in H , which is still H). $\square$

Remark

Cosets in general are not subgroups. In fact aH is a subgroup of H iff $a \in H$, so the only coset as a subgroup is H itself, generated by all aH where $a \in H$. (will elaborate on this later)

Exercise 4.3.4 (Cosets are complete partitions)

Show that any $g \in G$ must be in some coset of $H \leq G$. This means cosets of H can make up G , i.e.

$$H \cup aH \cup bH \cup \dots = G$$

(Hint: For all $g \in G$, g is in gH since $e \in H$.)

Theorem 4.3.5 (Coset Algebra)

Let H be a subgroup of G . Let $a, b \in G$. Then

$$aH = bH \iff H = a^{-1}bH$$

Proof. ($\implies$) Assume $aH = bH$. Then for all $h \in H$, we have $ah \in aH = bH$ and thus $ah = bh'$ for some $h' \in H$. Hence

$$h = a^{-1}bh' \in a^{-1}bH$$

and thus $H \subseteq a^{-1}bH$. Conversely, for all $a^{-1}bh \in a^{-1}bH$ where $h \in H$, since $bh \in bH$ and $aH = bH$, we know $bh \in aH$ and thus $bh = ah'$ for some $h' \in H$. Therefore,

$$a^{-1}bh = a^{-1}ah' = h' \in H$$

and then we have $a^{-1}bH \subseteq H$. We can conclude that $H = a^{-1}bH$.

($\Leftarrow$) Assume $H = a^{-1}bH$. For all $ah \in aH$ where $h \in H$, since $h \in H = a^{-1}bH$, we have $h = a^{-1}bh'$ for some $h' \in H$ and thus

$$ah = aa^{-1}bh' = bh' \in bH$$

Therefore, $aH \subseteq bH$. For all $bh \in bH$ where $h \in H$, we have $a^{-1}bh \in a^{-1}bH = H$ and thus $a^{-1}bh = h'$ for some $h' \in H$. Hence,

$$bh = ah' \in aH$$

and we then get $bH \subseteq aH$. We can conclude that $aH = bH$. $\square$

Remark

Note that $aH = bH$ does not necessarily imply $a = b$! Consider $2 + \mathbb{Z}_6 = 4 + \mathbb{Z}_6$ (we are using the additive notation here).

^aNote that this is an analogy to $ag = bg \iff g = a^{-1}bg$ for $g \in G$. It suggests that subgroup H has similar arithmetic rules as g does.

Corollary 4.3.6 (Key Lemma)

By 4.3.3 and 4.3.5, we have

$$aH = bH \iff a^{-1}b \in H$$

for $a, b \in G$ and $H \leq G$.

We show that any two cosets have the same size. Recall from B.2.6, $a + h$ for $h \in H, a \in G$ is a bijection from H to $a + H$. Since we mapped all elements in H to their corresponding and unique in $a + H$, the cosets should have the same size.

Theorem 4.3.7 (Cosets are equally sized)

Let G be finite and $H \leq G$. Then any two cosets of H have the same size. In particular,

$$|aH| = |H|, \quad \forall a \in G$$

Proof. For all $a \in G$, we can define the bijection (as in B.2.6) $f : H \mapsto aH$ by

$$f(h) = ah$$

with inverse $f^{-1} : aH \mapsto H$ defined by

$$f^{-1}(h) = a^{-1}h$$

By B.2.2, $|aH| = |H|$. $\square$

Lemma 4.3.8 (Cosets are non-overlapping)

Let G be finite and $H \leq G$. Any distinct cosets of H do not overlap.

Proof. We prove that if two cosets $aH \cap bH \neq \emptyset$ where $a, b \in G$ then $aH = bH$. Let $x \in aH \cap bH$, which exists.

Then $x = ah_1 = bh_2$ for some $h_1, h_2 \in H$. We have $h_1 = a^{-1}bh_2$ and thus $a^{-1}b = h_1h_2^{-1} \in H$. By 4.3.6, $aH = bH$. $\square$

Remark

Now we can make a connection to the previous remark. Since H , or equivalently all aH where $a \in H$, is a subgroup (also a coset) and contains e . However, 4.3.8 tells us that other cosets of H cannot have e , hence they cannot be a subgroup. Therefore, H (again, same as aH for all $a \in H$) is the only coset that is a subgroup.

The (most general) Lagrange's theorem can be proved very naturally with the above properties.

4.4 Proof of Lagrange's Theorem

Theorem 4.4.1 (Lagrange's Theorem)

Let G be finite and $H \leq G$ be a subgroup of G . Then $|H|$ divides $|G|$. Specifically, $\frac{|G|}{|H|}$ is the number of distinct cosets of H .

Proof. Let $a_1H, a_2H, \dots, a_kH$ be the k distinct cosets of H for $a_1, a_2, \dots, a_k \in G$. Note that one of them is H . By 4.3.4, they are complete such that

$$a_1H \cup a_2H \cup \dots \cup a_kH = G$$

By 4.3.8, they are disjoint such that

$$a_iH \cap a_jH = \emptyset, \quad \forall i \neq j \in \{1, \dots, k\}$$

By 4.3.7, they have the same size such that

$$|H| = |aH|, \quad \forall a \in G$$

Hence, by simple combinatorial tricks, we decompose G into k equally sized and disjoint subsets and count it by

$$\begin{aligned} |G| &= |a_1H \cup a_2H \cup \dots \cup a_kH| \\ &= |a_1H| + |a_2H| + \dots + |a_kH| \\ &= |H| + |H| + \dots + |H| \\ &= k|H| \end{aligned}$$

Therefore, $k = \frac{|G|}{|H|}$ is the number of distinct cosets of H . Clearly, $|H|$ divides $|G|$. $\square$

Inspired by the Lagrange's theorem, we can define the index for the number of cosets of a subgroup of G ,

Definition 4.4.2 (Index of a Subgroup)

The index of finite $H \leq G$, denoted $[G : H] = |G| / |H|$ which is a positive integer, is the number of distinct cosets of H in G .

For example, in 4.3.2,

$$[\mathbb{Z} : k\mathbb{Z}] = k \tag{4.4.1}$$

Corollary 4.4.3

If G is finite, then $[G : H]$ divides $|G|$. (Recall **Lagrange's Theorem** says $|G| = [G : H] \times |H|$.)

Corollary 4.4.4 (Lagrange's Theorem for Element Order)

If G is finite and if $a \in G$, then $|a|$ divides $|G|$. (This leads to $a^{|G|} = e$ by 1.4.13.)

Proof. By 4.2.3, $|a| = |\langle a \rangle|$. By 4.2.2, $\langle a \rangle$ is a subgroup of G . By **Lagrange's Theorem**, $|\langle a \rangle|$ divides $|G|$. $\square$

Corollary 4.4.5

If $|G| = p$ where p is prime then G must be cyclic.

Proof. Let $a \neq e \in G$ which exists since $p \geq 2$. By 4.4.4, $|a|$ divides $|G| = p$. The only case is $|a| = 1$ or p , and thus $|a| = p$ since $a \neq e$. By 4.2.7, $G = \langle a \rangle$ which is cyclic. (That said, if $|G|$ is prime then any non-identity element is a generator of G .) $\square$

5 Alternating Groups

Every $\sigma \in S_n$ can be written as a product of transpositions. For example, by (2.3.1),

$$(a_1 a_2 \cdots a_k) = (a_1 a_k) \cdots (a_1 a_3)(a_1 a_2)$$

We can see that a k -cycle can be written as a product of $k - 1$ transpositions. However, the 2-cycle factorization is not unique, e.g.

$$(a_1 a_2 \cdots a_k) = (a_1 a_2)(a_2 a_3) \cdots (a_{k-1} a_k)$$

is also a valid factorization. In general, the parity of number of transpositions is unique.

5.1 Definition

Theorem 5.1.1

Suppose $\sigma \in S_n$ has two distinct 2-cycle factorizations

$$\sigma = \alpha_1 \cdots \alpha_k = \beta_1 \cdots \beta_l$$

where α, β 's are transpositions. Then k and l are both even or both odd, i.e.

$$k \equiv l \pmod{2}$$

Proof. We claim that e can only be written as a product of even number of transpositions (show it). Note that

$$\begin{aligned} \sigma = \alpha_1 \cdots \alpha_k = \beta_1 \cdots \beta_l &\implies (\alpha_1 \cdots \alpha_k)^{-1} \beta_1 \cdots \beta_l = e \\ &\implies \alpha_k^{-1} \cdots \alpha_1^{-1} \beta_1 \cdots \beta_l = e \\ &\implies \underbrace{\alpha_k \cdots \alpha_1 \beta_1 \cdots \beta_l}_{k+l \text{ transpositions}} = e \end{aligned}$$

That said, $k + l$ is even. We thus have

$$k \equiv l \pmod{2}$$

□

Definition 5.1.2 (Sign of Permutation)

The sign of $\sigma \in S_n$ is defined by

$$\text{sign}(\sigma) = \begin{cases} +1 & \text{if } \sigma \text{ is a product of even number of transpositions} \\ -1 & \text{if } \sigma \text{ is a product of odd number of transpositions} \end{cases}$$

Remark

By (2.3.1), a k -cycle can be written as a product of $k - 1$ transpositions. Hence, a k -cycle is odd if k is even and is even if k is odd.

For example, $(12)(34)$ is even and $(1234) = (14)(13)(12)$ is odd.

Exercise 5.1.3 (Parity of Product of Permutations)

Let (*odd*) indicate an odd permutation and (*even*) indicate an even permutation. Show that

1. (*even*)(*even*) = (*even*)

$$2. (even)(odd) = (odd)$$

$$3. (odd)(even) = (odd)$$

This can be summarized as

$$\forall \sigma, \tau \in S_n, \quad \text{sign}(\sigma\tau) = \text{sign}(\sigma) \text{sign}(\tau)$$

Show that

$$\{\text{parity of } \sigma \mid \sigma \in S_n\} \cong (\{\pm 1\}, \times)$$

which is abelian.

Definition 5.1.4 (n -th Alternating Group)

The n -th alternating group is defined by

$$A_n = \{\sigma \in S_n \mid \sigma \text{ is even}\}$$

Similarly, the set of odd permutations is

$$O_n = S_n \setminus A_n = \{\sigma \in S_n \mid \sigma \text{ is odd}\}$$

It is also the only distinct coset of A_n .

5.2 Properties

Proposition 5.2.1

A_n is a subgroup of S_n .

Proof. By definition, $A_n \subseteq S_n$.

1. Since $\text{sign}(e) = 1$, we have $e \in A_n$ which is non-empty.

2. Let $\sigma, \tau \in A_n$. Then $\text{sign}(\sigma) = \text{sign}(\tau) = 1$. By 5.1.3,

$$\text{sign}(\sigma\tau) = \text{sign}(\sigma) \text{sign}(\tau) = 1$$

So $\sigma\tau \in A_n$.

3. Let $\sigma \in A_n$. By 5.1.3,

$$\begin{aligned} \text{sign}(e) = \text{sign}(\sigma\sigma^{-1}) = 1 &\implies \text{sign}(\sigma) \text{sign}(\sigma^{-1}) = 1 \\ &\implies 1 \cdot \text{sign}(\sigma^{-1}) = 1 \\ &\implies \text{sign}(\sigma^{-1}) = 1 \end{aligned}$$

Therefore, $\sigma^{-1} \in A_n$.

By 4.1.2, A_n is a subgroup of S_n . □

Exercise 5.2.2

Show that for all $\sigma \in S_n$,

$$\text{sign}(\sigma) = \text{sign}(\sigma^{-1})$$

(Hint: They are reciprocal to each other.)

Exercise 5.2.3

Is $(142)(23)(15)$ in A_5 ?

(Hint: Identify the sign of each cycle and apply 5.1.3. A poor method is write it as a product of transpositions and find the parity. The best way is to apply $\text{sign}(\cdot)$ on it and use the split property. You can also draw a Braid diagram and count the number of crossings, which has the same parity.)

Theorem 5.2.4 (Fundamental Theorem of Alternating Groups)

The only cosets of A_n are A_n itself and σA_n ^a where $\sigma \in S_n$ is odd. Therefore,

$$[S_n : A_n] = 2 \tag{5.2.1}$$

and

$$|A_n| = |S_n \setminus A_n| = \frac{n!}{2} \tag{5.2.2}$$

Proof. Consider σA_n where $\sigma \in S_n$. If $\sigma \in A_n$ then $\sigma A_n = A_n$ which is itself a coset, by 4.3.3. Now we prove that an odd coset is unique. Let $\sigma, \tau \in S_n \setminus A_n$ be any two odd permutations and we prove $\sigma A_n = \tau A_n$. By 4.3.6, we can just prove $\sigma^{-1}\tau \in A_n$. Note that $\text{sign}(\sigma) = \text{sign}(\tau) = -1$. By 5.1.3,

$$\begin{aligned} \text{sign}(\sigma^{-1}\tau) &= \text{sign}(\sigma^{-1}) \text{sign}(\tau) \\ &= \text{sign}(\sigma) \text{sign}(\tau) \text{ by 5.2.2} \\ &= -1 \cdot -1 \\ &= 1 \end{aligned}$$

Therefore, $\sigma^{-1}\tau \in A_n$ as desired. Following Lagrange's Theorem, we can naturally get those two equations. $\square$

^aIf σ is odd then σA_n is called an odd coset because its elements are odd. Try to show it by 5.1.3.

Exercise 5.2.5

Prove that $A_n \trianglelefteq S_n$.

(Hint: Show that $\sigma A_n \sigma^{-1} = A_n$, i.e. for any $c \in A_n$, $\sigma c \sigma^{-1}$ is still even for all $\sigma \in S_n$. You can apply $\text{sign}(\cdot)$ on $\sigma c \sigma^{-1}$ and use the split property.)

Exercise 5.2.6

Let $O_n = S_n \setminus A_n$ be the set of all odd permutations in S_n . Show that

$$S_n/A_n = \{A_n, O_n\}$$

By 1.4.8, $S_n/A_n \cong \mathbb{Z}_2$.

6 Homomorphisms

6.1 Definition

Definition 6.1.1 (Homomorphisms)

Let $(G, *)$, $(H, \cdot)$ be groups. A mapping $\phi : G \mapsto H$ is said to be a homomorphism if for all $x, y \in G$,

$$\phi(x * y) = \phi(x) \cdot \phi(y)^a$$

^aThat said ϕ "respects" the operation. I personally refer it to be the "split property".

The isomorphism (1.4.1) is indeed a bijective homomorphism.

Theorem 6.1.2 (Properties of the Homomorphism)

If $\phi : G \mapsto H$ is a homomorphism then for all $x \in G$,

1. $\phi(e_G) = e_H$
2. $\phi(x^{-1}) = \phi(x)^{-1}$
3. $\forall n \in \mathbb{Z}, \phi(x^n) = \phi(x)^n$
4. $|\phi(x)| \mid |x|$

Proof. Let $x \in G$.

1. Note that by split property of definition 6.1.1,

$$\phi(x) = \phi(x * e_G) = \phi(x) \cdot \phi(e_G) \implies \phi(e_G) = e_H$$

That said, ϕ always maps identity to identity.

2. Note that

$$\begin{aligned} \phi(e_G) &= \phi(x * x^{-1}) \\ &= \phi(x) \cdot \phi(x^{-1}) \text{ by split property} \end{aligned}$$

By property 1, $\phi(e_G) = e_H$ so

$$\phi(x) \cdot \phi(x^{-1}) = e_H \implies \phi(x^{-1}) = \phi(x)^{-1}$$

3. If $n = 0$ then clearly $\phi(e_G) = e_H$ by property. If $n > 0$ then by split property

$$\phi(x^n) = \underbrace{\phi(x) \cdots \phi(x)}_{n \text{ times}} = \phi(x)^n$$

If $n < 0$ then

$$\begin{aligned} \phi(x^n) &= \phi(x^{-1(-n)}) \\ &= \phi(x^{-1})^{-n} \text{ by split property} \\ &= \phi(x)^{-1(-n)} \text{ by property 2} \\ &= \phi(x)^n \end{aligned}$$

4. Note that

$$\begin{aligned} x^{|x|} = e_G &\implies \phi(x^{|x|}) = \phi(e_H) = e_G \text{ by 1} \\ &\implies \phi(x)^{|x|} = e_G \text{ by 3} \end{aligned}$$

By 1.4.13, $|\phi(x)|$ divides $|x|$.

Hence, a valid homomorphism must satisfy these 4 properties^a. □

^aNote that these are the fundamental properties of the homomorphism; they can be used for checking the validity of a homomorphism.

Note that property 4 of 6.1.2 does not require $|\phi(x)| = |x|$. Look at the following example,

Exercise 6.1.3

Let $\phi : (\mathbb{R}^n, +) \mapsto (\mathbb{R}^n, +)$ be defined by

$$\phi(x) = \mathbf{0}, \forall x \in \mathbb{R}^n$$

Prove that ϕ is a homomorphism.

Remark

A homomorphism $\phi : G \mapsto H$ is well defined if for $x, y \in G$,

$$x = y \implies \phi(x) = \phi(y)$$

For example,

$$f(x) = 3x : \mathbb{Z}_{10} \mapsto \mathbb{Z}_9$$

is not well defined since

$$f(1) = 1 \neq f(11) = 6$$

while $1 = 11$ in $\mathbb{Z}_{10}$. However,

$$f(x) = 3x : \mathbb{Z}_{10} \mapsto \mathbb{Z}_9$$

is well defined since

$$f(x) = 3x = f(x + 10m) = 3x + 30m$$

as $x = x + 10m$ and $3x = 3x + 30m$ in $\mathbb{Z}_{10}$.

6.2 Kernel and Image

Definition 6.2.1 (Kernel and Image of a Homomorphism)

Let $\phi : G \mapsto H$ be a homomorphism. Then the kernel and image are defined by

$$\text{Ker}(\phi) = \{g \in G \mid \phi(g) = e\}$$

and

$$\text{Img}(\phi) = \{\phi(g) \mid g \in G\}$$

Proposition 6.2.2

Let $\phi : G \mapsto H$ be a homomorphism. Then $\text{Ker}(\phi) \leq G$ and $\text{Img}(\phi) \leq H$.

Proof. We prove them one by one.

1. ($\text{Ker}(\phi) \leq G$): First note that $\text{Ker}(\phi) \subseteq G$. By 6.1.2, $\phi(e) = e$ so $e \in \text{Ker}(\phi)$, which is thus non-empty. Let $x, y \in \text{Ker}(\phi)$, then $\phi(x) = \phi(y) = e$. By split property 6.1.1,

$$\phi(xy) = \phi(x)\phi(y) = e$$

Therefore, $xy \in G$. Now let $x \in \text{Ker}(\phi)$. Again by 6.1.2,

$$\phi(x^{-1}) = \phi(x)^{-1} = e^{-1} = e$$

By 4.1.2, $\text{Ker}(\phi)$ is a subgroup of G .

2. ($\text{Img}(\phi) \leq H$): First note that $\text{Img}(\phi) \subseteq H$. By 6.1.2, $\phi(e) = e$ so $e \in \text{Img}(\phi)$, which is thus non-empty. Let $x, y \in \text{Img}(\phi)$. Then there exist $a, b \in G$ such that

$$\phi(a) = x \quad \phi(b) = y$$

By 6.1.1,

$$xy = \phi(a)\phi(b) = \phi(ab)$$

Since $ab \in G$, we have $xy = \phi(ab) \in \text{Img}(\phi)$. Now consider x^{-1} . By 6.1.2,

$$x^{-1} = \phi(a)^{-1} = \phi(a^{-1})$$

Since $a^{-1} \in G$, we have $x^{-1} = \phi(a^{-1}) \in \text{Img}(\phi)$. By 4.1.2, $\text{Img}(\phi)$ is a subgroup of H .

□

Exercise 6.2.3

Let ϕ be any homomorphism. Prove that $e \in \text{Ker}(\phi)$, that is, kernel of ϕ always contains e and is thus non-empty. (You might relate it to the nullspace of a matrix which always contains the zero vector.)

Proposition 6.2.4

A homomorphism $\phi : G \mapsto H$ is injective (B.1.1) if and only if $\text{Ker}(\phi) = \{e\}$.

Proof. ($\implies$) Since $\phi(e) = e$ and no two elements in G point to the same thing in H , the kernel should only contain e .

($\impliedby$) For the sake of contradiction, assume $\phi(x) = \phi(y)$ while $x \neq y$. Then

$$\phi(x)\phi(y)^{-1} = e \implies \phi(xy^{-1}) = e$$

Since $\text{Ker}(\phi) = \{e\}$, we must have

$$xy^{-1} = e \implies x = y$$

Contradiction. So ϕ is injective.

□

Exercise 6.2.5

Let $A \in \mathbb{R}^{m \times n}$. Show that if $Ax = b$ and $Ay = b$ for distinct $x, y \in \mathbb{R}^n$ and $b \in \mathbb{R}^m$, then $\text{Ker}(A) \neq \{0\}$.

6.3 Isomorphisms (Revised)

6.3.1 Definition

Definition 6.3.1 (Isomorphisms)

Let $\phi : G \mapsto H$ be a homomorphism. ϕ is an isomorphism if and only if it is bijective, i.e.

1. ϕ is injective, $\text{Ker}(\phi) = \{e\}$.
2. ϕ is surjective, $\text{Img}(\phi) = H$.

Remark

By B.2.3, ϕ is invertible, whose inverse $\phi^{-1} : H \mapsto G$ satisfying

$$\phi^{-1}(\phi(x)) = x, \forall x \in G$$

is also an isomorphism from H to G . (try to show it!)

Definition 6.3.2 (Group Isomorphisms)

Two groups G, H are isomorphic if there exists a well-defined isomorphism between them.

Remark

To prove two groups are isomorphic, it suffices to find an isomorphism between them. Remember to prove it's a bijective homomorphism (invertible!).

Exercise 6.3.3

Let $G = \langle a \rangle$ be a cyclic group of order n . Show that $G \cong \mathbb{Z}_n$.

(Hint: Define the mapping $\phi(a) = 1 : G \mapsto \mathbb{Z}_n$ and prove it's an isomorphism (i.e. a bijective homomorphism).)

Theorem 6.3.4

Let $\phi : G \mapsto H$ be an isomorphism (so $G \cong H$). Then

1. $|G| = |H|$.
2. G is abelian iff H is abelian.
3. $|x| = |\phi(x)|$.

Proof. We show that

1. Follows from B.2.2.
2. Assume G is abelian, i.e. $ab = ba$ for all $a, b \in G$. We show H is abelian. Let $x, y \in H$ be arbitrary. Since ϕ^{-1} is also an isomorphism^a, by split property 6.1.1,

$$\begin{aligned} \phi^{-1}(xy) &= \phi^{-1}(x)\phi^{-1}(y) \\ &= ab \text{ for some unique } a, b \in G \\ &= ba \text{ since } G \text{ is abelian} \\ &= \phi^{-1}(y)\phi^{-1}(x) \end{aligned}$$

$$= \phi^{-1}(yx)$$

We then have

$$\phi^{-1}(xy) = \phi^{-1}(yx) \implies \phi(\phi^{-1}(xy)) = \phi(\phi^{-1}(yx)) \implies xy = yx$$

So H is abelian. The converse can be proved in a similar way.

3. For $x \in G$, let $\phi(x) = b$ for unique $b \in H$. By 6.1.2, $|b| \mid |x|$. Since ϕ^{-1} is an isomorphism from H to G , $|\underbrace{\phi^{-1}(b)}_{=|x|}| \mid |b|$. Therefore, $|x| = |b|$.

□

^aOnly isomorphisms are invertible! Thus this property is not necessarily true for general homomorphisms.

Exercise 6.3.5

Show that

$$\phi(\text{Ker}(\phi)) = \{e\}$$

and

$$\phi(G) = H, \quad \phi^{-1}(H) = G$$

6.4 The First Isomorphism Theorem

We introduce a tool to construct an isomorphism.

Theorem 6.4.1 (The First Isomorphism Theorem (early version))

Let G, H be groups and let $\phi : G \mapsto H$ be an onto homomorphism. Let $K = \text{Ker}(\phi)$.

$$\begin{array}{ccc} G & & H \\ K & \mapsto & e \\ aK & \mapsto & a \\ bK & \mapsto & b \\ cK & \mapsto & c \\ & \dots & \end{array}$$

Then $[G : \text{Ker}(\phi)] = |\text{Img}(\phi)|$.

Proof. Recall that $\text{Ker}(\phi) \leq G$ (by 6.2.2) and $[G : \text{Ker}(\phi)] = m$ is the number of distinct left cosets of $\text{Ker}(\phi)$. Let $g \in G$. By 4.3.4, g is in one of the distinct left cosets

$$K, g_2K, \dots, g_mK$$

Suppose $g \in g_iK$ then $g = g_ik$ for some $k \in K$. Then

$$\begin{aligned} \text{Img}(\phi) &= \{\phi(g) \mid g \in G\} \\ &= \{\phi(g_ik) \mid g_i = e, g_2, \dots, g_m\} \\ &= \{\phi(g_i)\phi(k) \mid g_i = e, g_2, \dots, g_m\} \text{ by split property of } \phi \\ &= \{\phi(g_i) \mid g_i = e, g_2, \dots, g_m\} \text{ since } \phi(k) = e \\ &= \{e, \phi(g_2), \dots, \phi(g_m)\} \end{aligned}$$

Therefore,

$$|\text{Img}(\phi)| = m = [G : \text{Ker}(\phi)]$$

□

Definition 6.4.2 (Induced Isomorphisms)

Let $\phi : G \mapsto H$ be a homomorphism. Then

$$\psi : G / \text{Ker}(\phi) \rightarrow \text{Img}(\phi)$$

is the induced isomorphism.

For example, look at $\phi = \text{sign} : S_n \mapsto \{\pm 1\}$. Then

$$\text{Ker}(\phi) = A_n$$

and

$$S_n / \text{Ker}(\phi) = \{A_n, O_n\}$$

Clearly, this is isomorphic to the set

$$\{1, -1\}$$

Now look at $\phi(z) = z^n : \mathbb{C}^* \mapsto \mathbb{C}^*$. The kernel contains the n -th root of unity,

$$\text{Ker}(\phi) = U_n = \{1, e^{i2\pi/n}\}$$

Then

$$\mathbb{C}^* / U_n \cong \text{Img}(\phi) = \mathbb{C}^*$$

Theorem 6.4.3 (The First Isomorphism Theorem)

Let G, H be groups and let $\phi : G \mapsto H$ be an onto homomorphism. Let $K = \text{Ker}(\phi)$. Then

$$G / \text{Ker}(\phi) \cong \text{Img}(\phi)$$

with the induced isomorphism $\psi(gK) = \phi(g) : G/K \mapsto H$ (6.4.2). Thus $|G| = |\text{Img}(\phi)| |\text{Ker}(\phi)|$ if G, H are finite.

Proof. We first show that ψ is well defined, i.e.

$$\psi(gK) = \psi(g'K)$$

if $g = g'$. By split property,

$$\psi(gK) = \psi(g) \underbrace{\psi(K)}_{\{e\}} = \{\psi(g)\} = \{\psi(g')\} = \psi(g'K)$$

So it is well defined. We now show that $\psi(gK) = \phi(g) : G/K \mapsto H$ is an isomorphism by showing it's an onto and one-to-one homomorphism.

$$\psi(aK * bK) = \psi(abK) = \underbrace{\phi(ab) = \phi(a)\phi(b)}_{\text{by split property}} = \psi(aK)\psi(bK)$$

So ψ is a valid homomorphism. It is also invertible, and the inverse is $\psi^{-1}(\phi(g)) = gK : H \mapsto G/K$. It is well defined since if $\phi(g) = \phi(g')$,

$$\psi^{-1}(\phi(g)) = gK = g'K = \psi^{-1}(\phi(g'))$$

To show $gK \subseteq g'K$, let $gk \in gK$ and we show $gk \in g'K$. Note that by split property, $\phi(gk) = \phi(g)\phi(k) = \phi(g) = \phi(g')$, so $gk \in g'K$. We can similarly get $g'K \subseteq gK$. Thus $gK = g'K$. Now we have

$$\psi^{-1}(\psi(gK)) = gK$$

so it is the valid inverse. $\psi : G/H \mapsto H$ is thus an isomorphism.

Suppose $G/K = \{K, g_2K, \dots, g_mK\}$ is the quotient group (since the kernel K is normal) of G on $\text{Ker}(\phi)$. By **Lagrange's Theorem**, they are disjoint and

$$K \cup g_2K \cup \dots \cup g_mK = G$$

Hence

$$\begin{aligned} \text{Img}(\phi) &= \phi(G) \text{ by 6.3.5} \\ &= \phi(K \cup g_2K \cup \dots \cup g_mK) \\ &= \phi(K) \cup \phi(g_2K) \cup \dots \cup \phi(g_mK) \text{ since they are disjoint} \\ &= \phi(K) \cup \phi(g_2)\phi(K) \cup \dots \cup \phi(g_m)\phi(K) \text{ by split property of } \phi \\ &= \{e\} \cup \phi(g_2)\{e\} \cup \dots \cup \phi(g_m)\{e\} \text{ by 6.3.5} \\ &= \{e, \phi(g_2), \dots, \phi(g_m)\} \end{aligned}$$

We can clearly see its correspondence to $G/K = \{K, g_2K, \dots, g_mK\}$ through the isomorphism ψ . Therefore, $G/\text{Ker}(\phi) \cong \text{Img}(\phi)$. $\square$

Exercise 6.4.4

Show that $\mathbb{Z}_{10}/\langle 2 \rangle \cong \mathbb{Z}_2$. (Hint: Define the homomorphism $\phi(x) = x \pmod{2} : \mathbb{Z}_{10} \mapsto \mathbb{Z}_2$ which is onto. Then use the first isomorphism theorem.)

6.5 Classification of Finite Abelian Subgroups

Theorem 6.5.1

Let G be a finite abelian group. Then G is isomorphic to a direct product of cyclic groups such that

$$G \cong \mathbb{Z}_{a_1} \times \mathbb{Z}_{a_2} \times \dots \times \mathbb{Z}_{a_k}$$

where $|G| = a_1 a_2 \dots a_k$.

Theorem 6.5.2

Let $a, b \in \mathbb{N}^+$. If a, b are coprime then

$$\mathbb{Z}_a \times \mathbb{Z}_b \cong \mathbb{Z}_{ab}$$

7 Quotient Groups

7.1 Definition

In this section we look at the group of cosets.

Definition 7.1.1 (Quotient Set)

Let $H \leq G$. The quotient set of G on H , denoted G/H , is the set of all left cosets of H ,

$$\begin{aligned} G/H &= \{aH \mid a \in G\} \\ &= \{H, g_2H, \dots, g_mH\}, \quad g_2, \dots, g_m \in G \end{aligned} \quad (\text{distinct cosets})$$

where $m = [G : H]$ and each g_iH is a distinct coset of H .

Remark

By 4.3.8 and 4.3.4, the cosets are disjoint and complete. G/H is thus a decomposition of G . Furthermore, there can be many options for each $g_2, \dots, g_m$; for each g_iH of the m distinct cosets, g_i can be chosen to be any $a, b, c, \dots \in G$ such that $g_iH = aH = bH = cH = \dots$. By convention, g_1H is chosen to be the trivial coset H itself (4.3.1) with $g_1 \in H$. Note that any $a \in G$ must be in the coset aH since $e \in H$ (see 4.3.4), so for clearness we can reform it into

$$\begin{aligned} G/H &= \{K_1, K_2, \dots, K_m\}, \text{ all distinct cosets of } H \text{ and } K_1 = H \\ &= \{g_1H, g_2H, \dots, g_mH\} \\ &\text{for } g_1 \in K_1, g_2 \in K_2, \dots, g_m \in K_m \end{aligned} \quad (g\text{-form})$$

For example, we know $4\mathbb{Z} \leq \mathbb{Z}$. Then

$$\mathbb{Z}/4\mathbb{Z} = \{4\mathbb{Z}, 1 + 4\mathbb{Z}, 2 + 4\mathbb{Z}, 3 + 4\mathbb{Z}\}$$

and $\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}_4$.

Exercise 7.1.2

Let $H \leq G$. Suppose two cosets are equal, i.e. $aH = bH$ for some $a, b \in G$. Prove that $a = bg$ for some $g \in G$. (Hint: Use 4.3.6.)

Remark

In general, it does not strictly require $a = b$.

Define the operation on cosets of H to be

$$(aH) * (bH) = (ab)H, \quad a, b \in G \quad (7.1.1)$$

What constraints on H are needed for $(G/H, *)$ to be a valid group? We need the operation (7.1.1) to be well defined. For example, let

$$H = \{e, (12)\} \leq S_3$$

Then

$$(13)H = \{(13), (123)\} = (123)H, \quad (23)H = \{(23), (132)\} = (132)H$$

By (7.1.1),

$$(13)H * (12)H = (13)(12)H = (123)H$$

However,

$$(13)H * (12)H = (123)H * (132)H = (123)(132)H = H$$

They are not consistent. Thus $H = \{e, (12)\}$ is not suitable for this subgroup operation. To make (7.1.1) well defined, we need to make sure

$$aH * bH = (ab)H = (a'b')H = a'H * b'H$$

when $aH = a'H, bH = b'H$, that is, if $a' = ah_1$ and $b' = bh_2$ for some $h_1, h_2 \in H$ then $a'b' = (ah_1)(bh_2) = abh_3$. This needs

$$h_1bh_2 = bh_3 \implies h_1b = bh_3h_2^{-1} = bh_4$$

for some $h_4 \in H$, which implies

$$Hb = bH$$

Such H is called a normal subgroup.

Definition 7.1.3 (Normal Subgroups)

Let $H \leq G$. H is said to be a normal subgroup of G , denoted $H \trianglelefteq G$, if

$$aH = Ha$$

for all $a \in G$.

Remark

An abelian subgroup is always normal; however, a normal subgroup might not be abelian.

Proposition 7.1.4

If $H \trianglelefteq G$, i.e. $aH = Ha$ for all $a \in G$, if and only if

$$aHa^{-1} = H$$

Proof. ($\implies$):

($\subseteq$): Let $g \in aHa^{-1}$. Then $g = aha^{-1}$ for some $h \in H$. We want to show $g \in H$. Note that $ah \in aH = Ha$, so $ah = h'a$ for some $h' \in H$. Thus $g = aha^{-1} = h'aa^{-1} = h' \in H$.

($\supseteq$): Let $h \in H$. We want to show $h \in aHa^{-1}$. Since $ha \in Ha = aH$, $ha = ah'$ for some $h' \in H$ and thus $h = ah'a^{-1} \in aHa^{-1}$.

($\impliedby$): Exercise. □

Remark

aHa^{-1} is called the conjugation of h by a . H is normal iff it's closed under conjugation by all $a \in G$.

Definition 7.1.5 (Quotient Groups)

Let $H \trianglelefteq G$. $(G/H, *)$ is called the quotient group where

$$(aH) * (bH) = (ab)H$$

The identity is $eH = H$ and the inverse of $aH \in G/H$ is $(aH)^{-1} = a^{-1}H$. By **Lagrange's Theorem**, if G is finite then

$$|G/H| = [G : H] = |G| / |H|$$

Remark

The quotient group is a valid group. First, the operation is well defined on normal H , as shown before.

Proof. We prove the four axioms of a group.

- G/H is associative. Let $aH, bH, cH \in G/H$. Then

$$(aH * bH) * cH = (ab)H * cH = (abc)H$$

$$aH * (bH * cH) = aH * (bc)H = (abc)H$$

So $(aH * bH) * cH = aH * (bH * cH)$.

- Note that gH is a coset of H if $g \in G$. Let $aH, bH \in G/H$.

$$(aH) * (bH) = (ab)H$$

By closure property of G , $ab \in G$. So $(ab)H \in G/H$.

- Let $aH \in G/H$ be any coset of H . Then

$$(eH) * (aH) = (ea)H = aH$$

$$(aH) * (eH) = (ae)H = aH$$

So eH is a valid identity.

- Let $aH \in G/H$ be any coset of H . Then

$$(a^{-1}H) * (aH) = (a^{-1}a)H = eH$$

$$(aH) * (a^{-1}H) = (aa^{-1})H = eH$$

So $a^{-1}H$ is indeed the inverse of aH .

□

Remark

In quotient groups, the order of a coset refers to its order as an element in the quotient group, not its size (a coset might not be a group, so in general the size-order makes no sense on cosets).

If not specified, $N \trianglelefteq G$ is assumed for all quotient groups G/N .

Exercise 7.1.6 (Coset Operation Preserves Size)

Show that

$$|aN * bN| = |aN| = |bN| = |N|$$

(Hint: By 4.3.7, since the combination of two cosets is still a coset, the size is preserved.)

Exercise 7.1.7

Think about another operation $\star$ such that $(G/H, \star)$ is a valid group. You can modify the constraint on H , i.e. it doesn't have to be normal if you want.

Look at an example. Let $G = \mathbb{Z}_n \times \mathbb{Z}_m$ and $H = \mathbb{Z}_n \times \{0\}$. Since H is abelian, $H \trianglelefteq G$. The cosets of H are

$$\forall (a, b) \in G, \quad (a, b) + H = (0, b) + \underbrace{(a, 0) + H}_{=H} = (0, b) + H$$

That said,

$$G/H = \{(0, b) + H \mid b \in \mathbb{Z}_m\}$$

So $G/H \cong \mathbb{Z}_m$ by defining the isomorphism $(0, b) + H \in G/H \leftrightarrow b \in \mathbb{Z}_m$. This suggests that

$$\frac{\mathbb{Z}_n \times \mathbb{Z}_m}{\mathbb{Z}_n \times \{0\}} \cong \frac{\mathbb{Z}_m}{\{0\}} = \mathbb{Z}_m$$

It seems that the $\mathbb{Z}_n$ is "canceled out".

Proposition 7.1.8

Let $G = H \times K$ and both $\{e_H\} \times K, H \times \{e_K\}$ are normal. Then

$$\frac{H \times K}{H \times \{e_K\}} \cong K, \quad \frac{H \times K}{\{e_H\} \times K} \cong H$$

Proof. We prove the left one. First, the possible cosets of $H \times \{e_K\}$ are

$$\{(h, k)(H \times \{e_K\}) \mid (h, k) \in G \times K\} = \{H \times \{k\} \mid k \in K\}$$

By linking $H \times \{k\}$ to k in K , we can get an isomorphism between $\frac{H \times K}{H \times \{e_K\}}$ and K . It will be very similar to prove the right one. $\square$

Exercise 7.1.9

Let $\sigma \in S_n$. Prove that

$$\sigma(a_1 a_2 a_3 \cdots)(b_1 b_2 b_3 \cdots) \sigma^{-1} = (\sigma(a_1) \sigma(a_2) \sigma(a_3) \cdots)(\sigma(b_1) \sigma(b_2) \sigma(b_3) \cdots)$$

where $a_1, a_2, a_3, b_1, b_2, b_3, \dots \in S_n$ are distinct.

Remark

This shows that the conjugation on a permutation preserves its shape.

Exercise 7.1.10

Show that $\langle s \rangle = \{e, s\} \leq D_2$ is not normal. (Hint: $rsr^{-1} = r^2 s \neq \langle s \rangle$.)

Show that $\langle r \rangle = \{e, r\} \leq D_2$ is normal. (Hint: $r(r)r^{-1} = r$ is always a rotation, which is closed. Consider $sr s^{-1} = ss^{-1}r^{-1} = r^{-1}$. You flipped it twice, so it should be a rotation, not a reflection. It is also closed.)

Exercise 7.1.11

Inspired by the previous examples, prove that if $H \leq G$ and $[G : H] = 2$, then $H \trianglelefteq G$. (Hint: $G/H = \{H, aH\} = \{H, Ha\}$ where $a \notin H$.)

Theorem 7.1.12

Let $\phi : G \rightarrow H$ be a homomorphism. Then $\text{Ker}(\phi) \trianglelefteq G$.

Proof. Want to show $g \text{Ker}(\phi) g^{-1} = \text{Ker}(\phi)$ for all $g \in G$. Note that by split property of ϕ ,

$$\phi(g \text{Ker}(\phi) g^{-1}) = \phi(g) \phi(\text{Ker}(\phi)) \phi(g)^{-1} = e$$

Therefore, $g \text{Ker}(\phi) g^{-1} \subseteq \text{Ker}(\phi)$. Now let $k \in \text{Ker}(\phi)$. We want to show $k \in g \text{Ker}(\phi) g^{-1}$, i.e. $k = g k' g^{-1}$ for some $k' \in \text{Ker}(\phi)$. Let $k' = g^{-1} k g$, which is well defined and in the kernel, since $\phi(k') = \phi(g^{-1} k g) = \phi(g)^{-1} \phi(k) \phi(g) = e$ by split property. Since such k' exists, we have that $k \in g \text{Ker}(\phi) g^{-1}$. $\square$

Remark

In general, $\text{Img}(\phi) \leq H$ but might not be normal!

7.2 Towards The Correspondence Theorem**Exercise 7.2.1**

Let $N \trianglelefteq G$. Show that if $N \leq H \leq G$, then $N \trianglelefteq H$.

Theorem 7.2.2

Let $N \trianglelefteq G$. If $N \leq H \leq G$, then

$$H/N \leq G/N$$

Moreover, G/N is the grand set of all possible cosets of N under G and H/N takes some of them that are "related" to H , i.e. cosets of N under H . (will elaborate on this later)

Proof. Recall by definition 7.1.1, $H/N = \{hN \mid h \in H\}$ is a subset of $G/N = \{gN \mid g \in G\}$ since $H \subseteq G$. Since $N \in H/N$, it is non-empty. By 4.1.6, we show that $aN(bN)^{-1} \in H/N$ for all $aN, bN \in H/N$. Note that

$$aN(bN)^{-1} = aN(b^{-1}N) = ab^{-1}N \in H/N$$

since $ab^{-1} \in H$ which is a subgroup. Therefore $H/N \leq G/N$. □

Remark

The first isomorphism theorem has a more complete version for this theorem.

To visualize it, think about

$$\mathbb{Z}_8 = \{0, 1, 2, 3, 4, 5, 6, 7\} := G$$

and

$$\langle 2 \rangle = \{0, 2, 4, 6\} := H$$

with

$$\langle 4 \rangle = \{0, 4\} := N$$

Then

$$\begin{aligned} G/N &= \left\{ \overbrace{\{0, 4\}}^{e=N}, \overbrace{\{1, 5\}}^{N_2}, \overbrace{\{2, 6\}}^{N_3}, \overbrace{\{3, 7\}}^{N_4} \right\}, \text{ complete decomposition of } G \\ &= \{0 + \langle 4 \rangle, 1 + \langle 4 \rangle, 2 + \langle 4 \rangle, 3 + \langle 4 \rangle\} \\ &= \{4 + \langle 4 \rangle, 5 + \langle 4 \rangle, 6 + \langle 4 \rangle, 7 + \langle 4 \rangle\} \\ &= \{g_1 + \langle 4 \rangle, g_2 + \langle 4 \rangle, g_3 + \langle 4 \rangle, g_4 + \langle 4 \rangle\} \\ \text{where } g_1 &\in \underbrace{\{0, 4\}}_{e=N}, g_2 \in \underbrace{\{1, 5\}}_{N_2}, g_3 \in \underbrace{\{2, 6\}}_{N_3}, g_4 \in \underbrace{\{3, 7\}}_{N_4} \end{aligned}$$

Note the last line is the g -form we established in 7.1.1. G/N is the grand set, i.e. set of all possible cosets of N . Now look at

$$H/N = \left\{ \overbrace{\{0, 4\}}^{e=N}, \overbrace{\{2, 6\}}^{N_3} \right\}, \text{ complete decomposition of } H$$

$$= \{h_1 + \langle 4 \rangle, h_2 + \langle 4 \rangle\}, \quad h_1 \in \underbrace{\{0, 4\}}_{e=N}, h_2 \in \underbrace{\{2, 6\}}_{N_3}$$

Clearly H/N is a subset (also a subgroup) of G/N , since $h_1 h_2 \in H$ always. That is,

$$G/N = \{N, N_2, N_3, N_4\}$$

and

$$H/N = \{N, N_3\}$$

Here comes the question: How can we quickly find $N_i * N_j$, where $N_i, N_j \in G/N$? For example, $N_2 N_3$ is one of N, N_1, N_2, N_3 since G/N is a group. Since $N_2 = 1 + N, N_3 = 2 + N$, by definition of coset operation (7.1.1),

$$N_2 * N_3 = (1 + 2) + N = 3 + N = N_4$$

Or equivalently, only N_4 contains $(1+2) = 3$ so the product $N_2 * N_3$ must be N_4 . That said, if we find $N_i = g_i + N, N_j = g_j + N$, then $N_i N_j = N'$ where $g_i g_j \in N'$. Such N' is unique by 4.3.8. For convenience, we define the easier version of coset operation.

Definition 7.2.3 (Product of Group Subsets)

Let $H, K \subseteq G$. The product of H, K is defined by

$$HK = \bigcup_{h \in H} hK = \bigcup_{k \in K} Hk = \{hk \mid h \in H, k \in K\}$$

Remark

Note that if $F \subseteq H \leq G$ then $FH = H$, e.g. $HH = H$. Furthermore, $\{a\}K = aK, K\{a\} = Ka$ for any $a \in G, K \subseteq G$.

Exercise 7.2.4

Let $N \trianglelefteq G$. Then for any $H \subseteq G$, $HN = NH$.

Lemma 7.2.5 (Coset Construction)

Let $N \trianglelefteq G$. Suppose $[G : N] = m$ and

$$G/N = \{N_1, N_2, \dots, N_m\}, \quad N_1 = N$$

Then

$$N_i = gN, \quad \forall g \in N_i$$

Proof. Since $e \in N$, we must have $g \in gN$. By 4.3.8, gN must be the unique coset of N containing g , so $gN = N_i$ only. If g, g' are not from the same coset, then $gN \neq g'N$. Otherwise, if $gN = g'N = N_i$ for some coset N_i , then N_i contains both g, g' which is a contradiction. $\square$

Remark

This tells us that to see whether g, g' are in the same coset, always see whether $gN = g'N$. Hence, for any two $N_i, N_j \in G/N$, they can be written as

$$N_i = g_i N, N_j = g_j N, \quad \forall g_i \in N_i, g_j \in N_j \quad (7.2.1)$$

Lemma 7.2.6

Let $N \trianglelefteq G$. Suppose $[G : N] = m$ and

$$G/N = \{N_1, N_2, \dots, N_m\}, \quad N_1 = N$$

Then

$$N_i N_j \subseteq N_i * N_j$$

Proof. Since G/N is a well-defined quotient group, we have that $N_i * N_j = N_k$ for some coset $N_k \in G/N$. By (7.2.1), $g_i g_j N = N_k$ for all $g_i \in N_i, g_j \in N_j$. Since $e \in N$, $g_i g_j \in g_i g_j N = N_k$. $\square$

Theorem 7.2.7 (Coset Operation)

Let $N \trianglelefteq G$. Suppose $[G : N] = m$ and

$$G/N = \{N_1, N_2, \dots, N_m\}, \quad N_1 = N$$

Then for any $N_i, N_j \in G/N$,

$$N_i * N_j = N_i N_j$$

such that $N_i * N_j = N_k \in G/N$ is one of the cosets of N .

Proof. We know $N_i = g_i N, N_j = g_j N$ for some $g_i \in N_i, g_j \in N_j$. Then

$$\begin{aligned} N_i N_j &= (g_i N)(g_j N) \text{ by 7.2.5} \\ &= \{g_i n_1 g_j n_2 \mid \forall n_1, n_2 \in N\} \\ &= \{g_i n_1 n_3 g_j \mid \forall n_1, n_3 \in N\} \text{ since } g_j n_2 \in g_j N = N g_j \\ &= \{g_i n_4 g_j \mid \forall n_4 \in N\} \text{ for } n_4 = n_1 n_3 \in N \\ &= \{g_i g_j n_5 \mid \forall n_5 \in N\} \text{ for } n_5 \in N \text{ since } n_4 g_j \in N g_j = g_j N \\ &= g_i g_j N \\ &= (g_i N) * (g_j N) \\ &= N_i * N_j \end{aligned}$$

Another proof goes as

$$\begin{aligned} N_i N_j &= (g_i N)(g_j N) \text{ by 7.2.5} \\ &= (g_i N)(N g_j) \text{ by normality of } N \\ &= g_i N N g_j \\ &= g_i N g_j \text{ by associativity and } N N = N \\ &= g_i g_j N \text{ by normality of } N \\ &= (g_i N) * (g_j N) \\ &= N_i * N_j \end{aligned}$$

$\square$

Remark

Practically, it is enough to randomly pick some g_i, g_j and find which coset contains their product, since cosets are non-overlapping. (7.2.1) seems analogical to a subgroup $H \leq G$ such that $hH = H$ for all $h \in H$. We can now use $N_i N_j$ to represent $N_i * N_j$.

The result can be even more simplified.

Proposition 7.2.8 (Simplified Coset Operation)

Suppose $N_i N_j = N_k$ for $N_i, N_j, N_k \in G/N$. Then

$$g_i N_j = N_i g_j = N_k, \quad \forall g_i \in N_i, g_j \in N_j$$

Proof. By (7.2.5), let $N_i = g_i N$, $N_j = g_j N$ for some $g_i \in N_i, g_j \in N_j$. Note that

$$\begin{aligned} g_i N_j = N_k &\implies g_i g_j N = N_k \\ &\implies g_i g_j \in N_k \text{ since } e \in N \end{aligned}$$

So $g_i N_j$ and N_k are cosets that both contain $g_i g_j$. By 4.3.8, $g_i N_j = N_k$. We can similarly show that $N_i g_j = N_k$. $\square$

Corollary 7.2.9

Suppose $N_i N_j = N_k$ for $N_i, N_j, N_k \in G/N$. There are exactly $|N|$ permutations of $g_i g_j$ for $g_i \in N_i, g_j \in N_j$ such that $g_i g_j = g$.

Proof. By 7.2.8,

$$g_i N_j = N_i g_j = N_k$$

We look at $g_i N_j$ (symmetric to $N_i g_j$). By B.2.6, $g_i N_j$ is a bijection from N_j to N_k , so there exists a unique $g'_j \in N_j$ such that $g_i g'_j = g$ for each $g \in N_k$. $N_i N_j$ is the union of all such $g_i N_j$, i.e.

$$N_i N_j = \{g_i N_j \mid g_i \in N_i\} = \{g_{i,1} N_j, g_{i,2} N_j, \dots, g_{i,|N|} N_j\}$$

Each of them contributes one $g_i g_j$ that is equal to g , so there are $|N|$ such pairs in total. $\square$

Exercise 7.2.10

Use 7.2.6, 7.2.8 to prove 7.2.7. (Hint: By 7.2.8, $N_k \subseteq N_i N_j$, i.e. for each $g \in N_k$ there exists $g_i g_j = g$ for some $g_i \in N_i, g_j \in N_j$. With 7.2.6 such that $N_i N_j \subseteq N_k$, we can deduce that $N_i N_j = N_k$.)

Remark

We now have 7.2.6, 7.2.8 $\iff$ 7.2.7.

Exercise 7.2.11 (Coset Product Preserves Size)

Let G/N be a well-defined quotient group (i.e. $N \trianglelefteq G$). For any $N_i, N_j \in G/N$,

$$|N_i N_j| = |N_i| = |N_j| = |N|$$

This is analogical to 7.1.6. (Hint: Use 7.2.7.)

Now we can look into the example regarding the correspondence theorem.

$$\begin{aligned} G/H &= \{\{0, 2, 4, 6\}, \{1, 3, 5, 7\}\} \\ &= \{0 + \langle 2 \rangle, 1 + \langle 2 \rangle\} \cong \mathbb{Z}_2 \end{aligned}$$

and

$$\frac{G/N}{H/N} = \{N, N_2, N_3, N_4\} / \{N, N_3\} = \{\{N, N_3\}, \{N_2, N_4\}\} \cong G/H$$

7.3 The Correspondence Theorem

Corollary 7.3.1

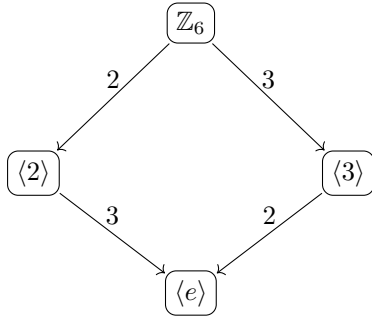
Let $K \leq H \leq G$ where G is finite. By **Lagrange's Theorem**,

$$[G : K] = [G : H][H : K]$$

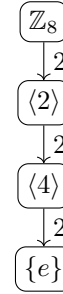
Proof. This is quite straightforward. By **Lagrange's Theorem**, $|H| = [H : K]|K|$, $|G| = [G : H]|H|$. Then

$$[G : K] = |G| / |K| = [G : H]|H| / |K| = [G : H][H : K]|K| / |K| = [G : H][H : K]$$

as desired. $\square$



(a) $\mathbb{Z}_6$ subgroup ancestor tree



(b) $\mathbb{Z}_8$ subgroup ancestor tree

Figure 7.1: Ancestor trees

For example, look at the tree of $\mathbb{Z}_6$. The edge numbers are the corresponding indices from each mother group to their subgroup. We can see that

$$[\mathbb{Z}_6 : \langle e \rangle] = [\mathbb{Z}_6 : \langle 2 \rangle][\langle 2 \rangle : \{e\}] = 3 \times 2 = 6$$

In the ancestor tree, how to find $[G : N]$ for any $N \leq G$? We can find any directed path from G to N and multiply the indices on the path edges. There can be multiple options for such G - N path.

Theorem 7.3.2 (The Correspondence Theorem (Lattice Isomorphism Theorem))

Let $N \trianglelefteq G$.

1. The functions α, β between $A = \{H \mid N \trianglelefteq H \leq G\}$ and $B = \{K \mid K \leq G/N\}$ by

$$\alpha(H) = H/N \stackrel{a}{:} A \mapsto B$$

$$\beta(K) = \{g \in G \mid gN \in K\} \stackrel{b}{:} B \mapsto A$$

are bijections.

Remark

Note that $G/N = \{N, g_2N, \dots, g_mN\}$ is the set of **ALL** distinct cosets of N with $m = [G : N]$, and $H/N = \{hN : h \in H\}$ is a subset of $G/N = \{gN : g \in G\}$ containing those cosets of N related to H . In this context, $K \leq G/N$ is a set of cosets of N , and $\beta(K)$ is the set of coefficients torn apart from the cosets in K (again, both g, g' are included if $gN = g'N$, though g might not be in $\{N, g_2N, \dots, g_mN\}$). All you need to know is, this theorem states that H/N gives a unique collection of cosets of N (from the grand set G/N), and the set of all coefficients (which is a subgroup of G) that reproduce these cosets is H .

2. If $H \trianglelefteq G$ such that $N \trianglelefteq H \trianglelefteq G$, then $H/N \trianglelefteq G/N$ and $(G/N)/(H/N) \cong G/H$.

Proof. Note N is normal, so the quotient groups are well-defined.

1. We need $\alpha \circ \beta, \beta \circ \alpha$ to be the identity function. Let $K \leq H/N$. We have

$$\begin{aligned}\alpha \circ \beta(K) &= \beta(K)/N \\ &= \{pN \mid p \in \beta(K)\} \\ &= \{pN \mid pN \in K\} \text{ since } \beta(K) = \{g \in G \mid gN \in K\} \\ &= K\end{aligned}$$

Let $H \in \{H \mid N \trianglelefteq H \leq G\}$.

$$\begin{aligned}\beta \circ \alpha(H) &= \{g \in G \mid gN \in \alpha(H)\} \\ &= \{g \in G \mid gN \in H/N\} \\ &= \{g \in G \mid g \in H\} \text{ since } H/N = \{hN \mid h \in H\} \\ &= H\end{aligned}$$

So α, β are well-defined bijections.

2. Let $N \trianglelefteq H \trianglelefteq G$. Then the quotient group G/H is well-defined. For convenience, we instead define

$$\begin{aligned}G/N &= \{N, N_2, \dots, N_m\}, \quad m = [G : N] \\ &= \{N, g_2N, \dots, g_mN\}, \quad g_2 \in N_2, \dots, g_m \in N_m\end{aligned}$$

and

$$\begin{aligned}H/N &= \{N_i \in G/N \mid N_i \subseteq H\}^c = \{N, N'_2, \dots, N'_p\} \subseteq G/N, \quad p = [H : N] \\ &= \{N, g'_2N, \dots, g'_pN\}, \quad g'_2 \in N'_2, \dots, g'_p \in N'_p\end{aligned}$$

By 7.2.2, $H/N \leq G/N$. We want to show that it's normal, i.e.

$$N_i(H/N)N_i^{-1} = H/N, \quad \forall N_i \in G/N$$

($\subseteq$): By (7.2.5), let $N_i = g_iN$ for $g_i \in N_i$, $N'_j = g'_jN$ for $g'_j \in N_j$, for any $N_i \in G/N, N'_j \in H/N$. Then

$$\begin{aligned}g_iN(g'_jN)g_i^{-1}N &= g'_jg_i g_i^{-1}N N N \text{ by normality of } N \\ &= g'_jN \in H/N\end{aligned}$$

So $N_i(H/N)N_i^{-1} \subseteq H/N$.

($\supseteq$): Let $g'_jN \in H/N$. For each $g_iN \in G/N$, define $g'_k = g_i^{-1}N(g'_jN)g_iN = g'_jN$ by normality of N and thus $g'_kN \in H/N$. Therefore, $g'_jN = g_iN(g'_kN)g_i^{-1}N \in N_i(H/N)N_i^{-1}$. So $N_i(H/N)N_i^{-1} \supseteq H/N$. We can thus deduce that $N_i(H/N)N_i^{-1} = H/N$ and H/N is normal.

We use **The First Isomorphism Theorem** to prove the second thing. We want an onto homomorphism $\phi : G/N \mapsto G/H$ whose kernel is H/N and image is G/H . It's time to define H 's cosets under G , i.e.

$$G/H = \{H, H_2, \dots, H_t\}, \quad t = [G : H]$$

Try to find some patterns - note both sequences of $H, H_2, \dots, H_t$ and $N, N_2, \dots, N_m$ can make up G . Some of the N_i 's produced under H , so we can fairly guess that each H_j would (disjointly) take the same amount of other N_i 's (see figure 7.2. this is referred to be the "different levels of partitions" of G). An option of ϕ is

$$\phi(gN) = gH^d \quad (7.3.1)$$

This is a homomorphism such that $\phi(N_i) = H_l$ if both N_i, H_l contain some $g \in G$. Note that

$$\phi(gN * g'N) = \phi(gg'N) = gg'H = gH * g'H = \phi(gN) * \phi(g'N)$$

Also $\text{Im}(\phi) = G/H$ since for each $gH \in G/H$, $\phi(gN) = gH$. As well, $\text{Ker}(\phi) = H/N$ since the solution to $\phi(gN) = H$ is gN for any $g \in H$, which is H/N . By the first isomorphism theorem,

$$(G/N)/(H/N) \cong G/H$$

□

Remark

At this stage you might have noticed that we are playing with two sets of group systems:

$$N \leq H \leq G$$

and

$$H/N \leq G/N$$

whose elements are cosets.

^aRecall $H/N \leq G/N$ by 7.2.2.

^bThis is a subgroup of G . First, identity $N \in K$ since $K \leq G/N$ so $\beta(K)$ is non-empty. Then let any $a, b \in \beta(K)$ such that $aN, bN \in K$. We show that $ab^{-1} \in \beta(K)$. Note that $ab^{-1}N = aN * b^{-1}N = aN * (bN)^{-1} \in K$ since K is a subgroup. By 4.1.6, $\beta(K) \leq G$.

^cLike stated before, H/N takes some cosets in G/N that are related to H . Also note that cosets in H/N can be unionized to make up H (by 4.3.4), so it makes sense for it to take those cosets that are subsets of H .

^dYou can also write $\phi(N_i) = H_l$ where $H_l \in G/H$ is the unique mother set such that $N_i \subseteq H_l$, but it's harder to prove.

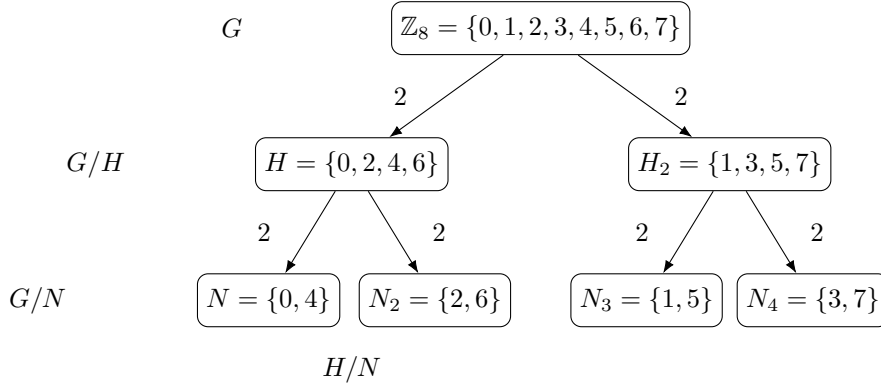


Figure 7.2: Partition Tree of $\mathbb{Z}_8$

It seems that the isomorphism for $(G/N)/(H/N) \cong G/H$ is $H_l \leftrightarrow \{N_i \mid N_i \subseteq H_l\}$ iff $H_l = \bigcup \{N_i \mid N_i \subseteq H_l\}$. You can also prove that the partition diagram is a tree, i.e. it contains no cycles. A cycle is like, for example, there exist $g, g' \in N_i$ such that $g \in H_l, g' \in H_k$ for distinct cosets H_l, H_k (this means each coset of N is completely in some coset of H).

8 Group Actions

We have seen many examples of group actions. For instance, the linear mapping

$$L(x) = Ax : \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad A \in GL_3(\mathbb{R})$$

is a group action such that A acts on x , where A transforms x to some vector in $\mathbb{R}^3$. The dihedral group also consists of all possible permutations of rotations and reflections that can act on the dihedral n -gon.

8.1 Definition

Definition 8.1.1 (Group Actions)

Let G be a group and X a non-empty set. We say G acts on X , denoted $G \curvearrowright X$, via the group action which is a well-defined function $f : G \times X \mapsto X$ where

$$f(g, x) = gx$$

that satisfies

$$ex = x, \quad g(hx) = (gh)x$$

for all $g, h \in G$.

Remark

The choice of X should be consistent with G 's elements. G can also act on itself, which is very similar to GG in 7.2.3.

Definition 8.1.2 (Orbits and Stabilizers)

Suppose $G \curvearrowright X$. Let $x \in X$.

- The orbit of x is the range of the group action, i.e.

$$\text{Orb}(x) = \{gx \mid g \in G\} \subseteq X$$

- The stabilizer of x is the set of elements in G that preserve x , i.e.

$$\text{Stab}(x) = \{g \in G \mid gx = x\}$$

Exercise 8.1.3 (Orbits Are Equally Sized)

Suppose $G \curvearrowright X$. Let $x, x' \in X$. Show that $|\text{Orb}(x)| = |\text{Orb}(x')|$. (Hint: Find a bijection between them.)

Theorem 8.1.4

Suppose $G \curvearrowright X$. Let $x \in X$. Then $\text{Stab}(x) \leq G$.

Proof. Note that $e \in \text{Stab}(x)$, so $\text{Stab}(x)$ is non-empty. Let $g, h \in G$, then $gx = x, hx = x$. Then $gh^{-1}x = gh^{-1}hx = gx = x$ so $gh^{-1} \in \text{Stab}(x)$. It is thus a subgroup of G by 4.1.6. $\square$

Remark

In general, $\text{Orb}(x)$ is not a group. It's a group only when it's onto.

Theorem 8.1.5 (Orbit-stabilizer Theorem)

Suppose $G \curvearrowright X$. Then for all $x \in G$,

$$|G| = |\text{Orb}(x)| |\text{Stab}(x)|$$

Proof. We want to show $|G/\text{Stab}(x)|^a = |\text{Orb}(x)|$. Let $S = \text{Stab}(x)$. We claim that

$$\phi(gS) = gx : G/\text{Stab}(x) \mapsto \text{Orb}(x)$$

is a bijection. It is well-defined since if $g = g' \in G$,

$$\phi(gS) = gx = g'x = \phi(g'S)$$

It is invertible. The inverse is

$$\phi^{-1}(gx) = gS : \text{Orb}(x) \mapsto G/\text{Stab}(x)$$

The inverse is also well defined. Note that

$$\phi^{-1}(\phi(gS)) = \phi^{-1}(gx) = gS$$

and

$$\phi(\phi^{-1}(gx)) = \phi(gS) = gx$$

□

^aSince $\text{Stab}(x)$ is not necessarily normal so it just means the set of left cosets.

For example, let $G = S_n$ and $X = \{1, \dots, n\}$. Then

$$\text{Stab}(1) = \{\text{permutations of } 2, \dots, n\} \rightarrow (n-1)! \text{ elements}$$

and

$$\text{Orb}(1) = \{1, \dots, n\} \rightarrow n \text{ elements}$$

Clearly,

$$|\text{Stab}(1)| |\text{Orb}(1)| = (n-1)!n = n! = |S_n|$$

Proposition 8.1.6

Orbits are either disjoint or equal. They cover all $x \in X$. (This is analogical to 4.3.8 and 4.3.4.)

Proof. Orbits cover all $x \in X$ since $x = ex \in \text{Orb}(x)$. Now let $\text{Orb}(x), \text{Orb}(x')$ be orbits on $x, x' \in X$ respectively. Clearly $\text{Orb}(x) = \text{Orb}(x')$ when $x = x'$. Now suppose $x \neq x'$ but $\text{Orb}(x) \cap \text{Orb}(x') \neq \emptyset$. Let $gx = g'x'$ for some $g, g' \in G$, then $x = \underbrace{g^{-1}g'}_{\in G} x' \in \text{Orb}(x')$. Hence,

$$\text{Orb}(x) = \{hx \mid h \in G\} = \{hg^{-1}g'x' \mid h \in G\} = \{h'x' \mid h' \in G\} = \text{Orb}(x')$$

Recall that $hg^{-1}g'$ is some unique h' in G , by similar idea in B.2.6. This shows that if $x \in \text{Orb}(x')$ then $\text{Orb}(x) = \text{Orb}(x')$. Thus, we can conclude that if $\text{Orb}(x)$ and $\text{Orb}(x')$ for $x, x' \in X$ are overlapping then they must be equal. Also note that it's possible for them to be disjoint. □

Remark

By 8.1.3 and 8.1.6, distinct orbits partition X . This is analogical to Lagrange's Theorem.

Theorem 8.1.7 (Cauchy's Theorem)

Let G be a finite group. If a prime p divides $|G|$, then G must contain an element of order p .

8.2 Burnside's Lemma**Definition 8.2.1** (Fixes)

The fix of $g \in G$ in X is defined by

$$\text{Fix}(g) = \{x \in X \mid gx = x\} \subseteq X$$

Proposition 8.2.2

Let $x \in X$ be arbitrary. Let $O_x = \{x_1, x_2, \dots, x_n\} \subseteq X$ be those of the same orbit as x , i.e.

$$\text{Orb}(x) = \text{Orb}(x_1) = \text{Orb}(x_2) = \dots = \text{Orb}(x_n)$$

Then

$$\sum_{h \in O_x} \frac{1}{|\text{Orb}(h)|} = 1 \quad (1)$$

We can then conclude that

$$\sum_{x \in X} \frac{1}{|\text{Orb}(x)|} = N \quad (2)$$

where N is the number of distinct orbits in X .

Proof. We just want to prove $|O_x| = |\text{Orb}(x)|$. By 8.1.6, the distinct orbits partition X , so let

$$O_1, \dots, O_N \quad (\star)$$

be the N distinct orbits in X . Similar to (7.2.5), for any $y_1, y_2 \in X$,

$$O_i = \text{Orb}(y_1) = \text{Orb}(y_2)$$

if and only if both $y_1, y_2 \in O_i$ since $y_1 \in \text{Orb}(y_1), y_2 \in \text{Orb}(y_2)$. So what is O_x ? Suppose x is in one of the distinct orbits in $(\star)$, say O_i , so $\text{Orb}(x) = O_i$. From the last equation we see that $\text{Orb}(x) = \text{Orb}(y)$ for all $y \in O_i$, so $O_x = O_i$. Recall that by 8.1.3, orbits have the same size which is $|\text{Orb}(x)|$, we have

$$|O_x| = |O_i| = |\text{Orb}(x)|$$

We thus get equation (1). To get equation (2), note that

$$\sum_{x \in X} \frac{1}{|\text{Orb}(x)|} = \sum_{i=1}^N \sum_{h \in O_i} \frac{1}{|\text{Orb}(h)|} = \sum_{i=1}^N 1 = N$$

as desired. □

Lemma 8.2.3 (Burnside's Lemma)

Let G be finite and $G \curvearrowright X$. Let N be the number of orbits in X . Then

$$N = \frac{1}{|G|} \sum_{g \in G} \text{Fix}(g)$$

Proof. Define

$$Y = \{(g, x) \mid gx = x, g \in G, x \in X\}$$

We claim that

$$|Y| = \sum_{g \in G} \text{Fix}(g)$$

and

$$\begin{aligned} |Y| &= \sum_{x \in X} \text{Stab}(x) \\ &= \sum_{x \in X} \frac{|G|}{|\text{Orb}(x)|} \text{ by 8.1.5} \\ &= |G| \sum_{x \in X} \frac{1}{|\text{Orb}(x)|} \\ &= |G| N \text{ by 8.2.2} \end{aligned}$$

Combining them, we get

$$|Y| = \sum_{g \in G} \text{Fix}(g) = |G| N$$

so we can naturally get

$$N = \frac{1}{|G|} \sum_{g \in G} \text{Fix}(g)$$

as desired. □

A Number Theory

A.1 Divisibility

Theorem A.1.1 (Remainder Theorem)

Let $n \geq 0, m > 0 \in \mathbb{Z}$. There exist unique $k, r \in \mathbb{Z}$ such that $n = km + r$ and $0 \leq r < m$.

Proof. Fix $m > 0 \in \mathbb{Z}$. We prove by induction on n .

Base Case When $n = 0$, we have $n = 0m + 0$.

Inductive Hypothesis For $n \geq 0$, $\exists! k, r$ such that $n = km + r$ and $0 \leq r < m$.

Inductive Steps We wish to prove for $n + 1$. We have

$$n + 1 = km + r + 1$$

Note that $r + 1 \leq m$ since $r < m$. We have two cases.

Case $r + 1 < m$: Let $r' = r + 1$ which is unique and $r' < m$. Hence $n + 1 = km + r'$.

Case $r + 1 = m$: Then

$$n + 1 = km + r + 1 = (k + 1)m + 0 = k'm + r'$$

where $k' = k + 1$ and $r' = 0$.

Conclusion Since we proved that $\exists! k', r'$ such that $n + 1 = k'm + r'$ and $0 \leq r' < m$, by induction, all $n \geq 0 \in \mathbb{Z}$ can be written as $n = km + r$ for some integer k and $0 \leq r < m$. $\square$

A.2 Greatest Common Divisors

Corollary A.2.1

Let $a, b \in \mathbb{Z}$. Then $\gcd(a, b) \mid a$ and $\gcd(a, b) \mid b$.

Lemma A.2.2

Let $a, b, k \in \mathbb{Z}$. Then

$$\gcd(ak, bk) = |k| \gcd(a, b)$$

Lemma A.2.3

Let $a, b \in \mathbb{Z}$. Let $g = \gcd(a, b)$. Then

$$\gcd\left(\frac{a}{g}, \frac{b}{g}\right) = 1$$

Lemma A.2.4

Let $a, b, k \in \mathbb{Z}$. If $k \mid a$ and $k \mid b$, then $k \mid \gcd(a, b)$.

A.3 Linear Diophantine Equations

Lemma A.3.1 (Bezout's Lemma)

Let $a, b \in \mathbb{Z}$. Then there exist $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a, b)$$

Theorem A.3.2 (Solvability of LDE)

Let $a, b, d \in \mathbb{Z}$ and $a, b \neq 0$. The equation

$$ax + by = d$$

has an integer solution (x, y) if and only if $\gcd(a, b) \mid d$.

B Mapping

B.1 Surjective and Injective Functions

Definition B.1.1 (Injective Functions)

A function $f : A \mapsto B$ is one-to-one (injective) if

$$\forall x, y \in A, \quad x \neq y \implies f(x) \neq f(y) \text{ or equivalently } f(x) = f(y) \implies x = y$$

A 1-1 function $f : A \mapsto B$ maps two distinct elements in domain A to two distinct elements in co-domain B . That is, if x, y are different, then $f(x), f(y)$ must be different. Informally, I say $x \in A$ points to $y \in B$ via f if $f(x) = y$. Note that f does not need to point to all elements in B but $f(x)$ must be well defined and exist in B for all $x \in A$, that is, $f(A) \subset B$. Also notice that $|A| \leq |B|$.

Exercise B.1.2

Let $f : \mathbb{R}^{-a} \mapsto \mathbb{C}$ be defined by

$$f(x) = \sqrt{x}$$

Show that f is injective.

^aIn this question, $\mathbb{R}^- = \{x \in \mathbb{R} \mid x \leq 0\}$

Definition B.1.3 (Surjective Functions)

A function $f : A \mapsto B$ is onto (surjective) if

$$\forall y \in B, \quad \exists x \in A \text{ such that } f(x) = y$$

An onto function must be able to point to all elements in B using elements in A . However, there can be duplicate pointing ($f(x) = f(y)$ for $x \neq y$), as long as all elements in B are pointed to. In this case, $|A| \geq |B|$.

Exercise B.1.4

Let $f : \mathbb{R} \mapsto \mathbb{R}^{+a}$ be defined by

$$f(x) = x^2$$

Show that f is surjective.

^aIn this question, $\mathbb{R}^+ = \{x \in \mathbb{R} \mid x \geq 0\}$

B.2 Bijections

Definition B.2.1 (Bijections)

A function $f : A \mapsto B$ is a bijection if it is both onto and 1-1.

Corollary B.2.2

If $f : A \mapsto B$ is a bijection, then $|A| = |B|$.

Proof. By definition, f is onto, so $|A| \geq |B|$. f is also 1-1, so $|A| \leq |B|$. Therefore $|A| = |B|$. □

Lemma B.2.3 (Invertibility of Bijections)

A function $f : A \mapsto B$ is bijective if and only if it is invertible. That is, for all $y \in B$, there exists a unique $x \in A$ such that $f(x) = y$. The inverse is denoted $f^{-1} : B \mapsto A$ where $f^{-1}(f(x)) = f^{-1}(y) = x$.

Proof. (bijective $\implies$ invertible) By definition of onto functions, there exists $x \in A$ such that $f(x) = y$ for all $y \in B$. Now assume a contradiction that there are at least two different $x, z \in A$ such that $f(x) = f(z) = y$. By definition of 1-1 functions, if $f(x) = f(z)$ then $x \neq y$. Contradiction. So $\forall y \in B, \exists! x \in A$ such that $f(x) = y$. Record each pair of such (x, y) (all pairs are different) and define $f^{-1}(y) = x$ for each pair of them. We have $f(f^{-1}(y)) = y$ and $f^{-1}(f(x)) = x$, and thus f^{-1} is a valid inverse of f .

(invertible $\implies$ bijective) We first prove it is onto. For all $y \in B$, let $x = f^{-1}(y)$ which exists in A . Then $f(x) = f(f^{-1}(y)) = y$, which means f is onto. Now we prove it is 1-1. Assume a contradiction that $x \neq x'$ but $f(x) = f(x')$. Let $y = f(x) = f(x')$, but now $f^{-1}(y) = x$ and $f^{-1}(y) = x'$. Since $x \neq x'$, this is a contradiction. Therefore, f is 1-1 and thus bijective. $\square$

A proof in plain language is: a bijection is onto, so for all $y \in B$ there must be some $x \in A$ pointing to y ; a bijection is also 1-1, so there will only be one $x \in A$ pointing to y , if any. Therefore the inverse of y exists, which is the unique x . If f is invertible and $f(x) = y$, then we say there exists a unique correspondence between x and y , and thus there exists an isomorphism between A and B .

Exercise B.2.4

Note that given a function $f : A \mapsto B$ and a subset $S \subseteq A$, the subrange of f on S is defined by

$$f(S) = \{f(s) \mid s \in S\} \subseteq B$$

Prove that if f is a bijection then

$$f(A) = B$$

and

$$f^{-1}(B) = A$$

For example, we define the linear mapping

$$f : \mathbb{R}^3 \mapsto \mathbb{R}^3, \quad f(x) = Ax$$

where $A \in \mathbb{R}^{3 \times 3}$ is invertible. If $A = I$, then $f(x) = x$ and it is an identity mapping. Otherwise, A maps x to some other vector y in $\mathbb{R}^3$. The inverse of $f : \mathbb{R}^3 \mapsto \mathbb{R}^3$ is thus $f^{-1}(y) = A^{-1}y$ for all $y \in \mathbb{R}^3$.

As another example, consider

$$f : \mathbb{R}^4 \mapsto \mathbb{P}^4, \quad f(x_1, x_2, x_3, x_4) = 2x_1 + 3x_2z + 4x_3z^2 + 5x_4z^3$$

where $\mathbb{P}^4$ is the polynomial space of degree 4. As you can see, a bijection's domain A and co-domain B don't have to be the same, just need a unique relationship between their elements. The inverse function is thus

$$f^{-1} : \mathbb{P}^4 \mapsto \mathbb{R}^4, \quad f^{-1}(x_1 + x_2z + x_3z^2 + x_4z^3) = \left(\frac{x_1}{2}, \frac{x_2}{3}, \frac{x_3}{4}, \frac{x_4}{5}\right)$$

Now consider a more straightforward example. Let $f : \{1, 2, 3\} \mapsto \{1, 2, 3\}$ have the following mapping

$$\begin{array}{ccc} 1 & \xrightarrow{f} & 3 \\ 2 & \xrightarrow{f} & 1 \\ 3 & \xrightarrow{f} & 2 \end{array}$$

Then the inverse $f^{-1} : \{1, 2, 3\} \mapsto \{1, 2, 3\}$ follows

$$\begin{array}{ccc} 1 & \xrightarrow{f^{-1}} & 2 \\ 2 & \xrightarrow{f^{-1}} & 3 \\ 3 & \xrightarrow{f^{-1}} & 1 \end{array}$$

where you can verify that $f^{-1}(f(x)) = x$ for all $x \in \{1, 2, 3\}$.

Exercise B.2.5

Find a bijection to prove

$$\binom{n}{k} = \binom{n}{n-k}$$

Do not use the combination formula!

(Hint: In $\{1, 2, \dots, n\}$, the number of ways of picking k objects is the same as that of picking the rest $n - k$ objects. If we let A be the set consisting of all k -subsets and B be the set consisting of all $(n - k)$ -subsets, what can a bijection $f : A \mapsto B$ be? It will then be sufficient to show $|A| = |B|$.)

Exercise B.2.6 (Group Bijection)

Let $g \in G$ where G is a group. Show that $f(x) = gx : G \mapsto G$ is a bijection and find its inverse.

Exercise B.2.7

Let $f : \mathbb{R}^+ \mapsto \mathbb{R}^{+^a}$ be defined by

$$f(x) = x^2$$

Show that f is bijective and find its inverse.

^aIn this question, $\mathbb{R}^+ = \{x \in \mathbb{R} \mid x \geq 0\}$

Exercise B.2.8

Suppose $f(x) = Ax : \mathbb{R}^3 \mapsto \mathbb{R}^3$ and $A \in \mathbb{R}^{3 \times 3}$ has determinant 0. Explain why this function is not a bijection.

Exercise B.2.9

Let $f(x(t)) = \frac{d}{dt}x(t) : C^\infty \mapsto C^\infty{}^a$. Prove or disprove that f is a bijection.

^a C^∞ is the set of all differentiable functions.

Exercise B.2.10 (Closure of Bijection Composition)

Let $f : A \mapsto B, g : B \mapsto C$ be bijections. Show that $g \circ f : A \mapsto C$ is a bijection.

Exercise B.2.11 (Associativity of Bijection Composition)

Let $f : A \mapsto B, g : B \mapsto C, h : C \mapsto D$ be bijections. Show that $(h \circ g) \circ f = h \circ (g \circ f)$. (Hint: How would you prove the equivalence of two functions?)

C Indices

List of Theorems

1.1.1 Definition (Groups)	4
1.1.2 Definition (Abelian Groups)	5
Remark	5
1.2.1 Exercise	6
1.3.1 Exercise (Groups of Linear Spaces)	6
1.3.2 Exercise (The Unit Circle)	7
1.3.3 Exercise (Element-wise Vector Product)	7
1.3.4 Exercise (Direct Product)	7
1.3.5 Exercise (XOR Group)	7
1.3.6 Exercise (Group for Symmetric Difference)	8
1.3.7 Exercise	8
1.3.8 Exercise	8
1.4.1 Lemma (Uniqueness of Identity)	9
1.4.2 Exercise	9
1.4.3 Lemma (Uniqueness of Inverse)	9
1.4.4 Lemma	9
1.4.5 Lemma (Inverse of a Product of Group elements)	9
1.4.6 Proposition	10
1.4.7 Example	11
Solution (Shortened)	11
1.4.8 Exercise	11
1.4.9 Exercise	11
1.4.10 Definition (Group Order)	11
1.4.11 Definition (Element Order)	11
1.4.12 Proposition	12
1.4.13 Proposition	12
1.4.14 Exercise	12
2.2.1 Example	14
2.3.1 Definition (k -cycle)	16
2.3.2 Definition (Transpositions)	16
2.3.3 Proposition	16
2.4.1 Lemma (Order of k -Cycles)	17
2.4.2 Proposition	18
2.4.3 Exercise	19
Remark	19
2.5.1 Exercise	20
2.5.2 Exercise	21
3.1.1 Exercise	23
3.1.2 Exercise	23
3.2.1 Exercise	23
3.2.2 Exercise	23
4.1.1 Definition (Subgroups)	24
4.1.2 Lemma (Subgroup Test)	24
4.1.3 Corollary	25
4.1.4 Example	25
4.1.5 Exercise	25
4.1.6 Lemma (Condensed Subgroup Test)	26

4.1.7 Exercise	26
4.2.1 Definition (Cyclic Groups)	27
4.2.2 Exercise (Subgroup Generators)	27
4.2.3 Exercise	27
4.2.4 Exercise (Cyclic Groups are Abelian)	28
4.2.5 Exercise	28
4.2.6 Exercise	28
4.2.7 Proposition	28
4.2.8 Theorem	29
4.2.9 Proposition	30
Remark	30
4.2.10 Corollary (Counting Generators of a Cyclic Group)	30
4.2.11 Example	31
4.2.12 Proposition	31
Remark	31
4.2.13 Corollary	31
4.2.14 Exercise (Counting Distinct subgroups of a Cyclic Group)	31
4.2.15 Proposition	32
4.2.16 Exercise	32
4.2.17 Theorem (Lagrange's Theorem for Cyclic Groups)	32
4.2.18 Corollary (Lagrange's Theorem for Element Order (Cyclic Groups))	33
4.2.19 Exercise (Lagrange's Theorem for Abelian Groups)	33
Remark (Aside)	33
4.2.20 Exercise	33
4.3.1 Definition (Cosets)	34
Remark	34
Remark	34
4.3.2 Exercise	35
4.3.3 Proposition	35
Remark	35
4.3.4 Exercise (Cosets are complete partitions)	35
4.3.5 Theorem (Coset Algebra)	35
Remark	36
4.3.6 Corollary (Key Lemma)	36
4.3.7 Theorem (Cosets are equally sized)	36
4.3.8 Lemma (Cosets are non-overlapping)	36
Remark	37
4.4.1 Theorem (Lagrange's Theorem)	37
4.4.2 Definition (Index of a Subgroup)	37
4.4.3 Corollary	37
4.4.4 Corollary (Lagrange's Theorem for Element Order)	38
4.4.5 Corollary	38
5.1.1 Theorem	39
5.1.2 Definition (Sign of Permutation)	39
Remark	39
5.1.3 Exercise (Parity of Product of Permutations)	39
5.1.4 Definition (n -th Alternating Group)	40
5.2.1 Proposition	40
5.2.2 Exercise	40
5.2.3 Exercise	41
5.2.4 Theorem (Fundamental Theorem of Alternating Groups)	41

5.2.5 Exercise	41
5.2.6 Exercise	41
6.1.1 Definition (Homomorphisms)	42
6.1.2 Theorem (Properties of the Homomorphism)	42
6.1.3 Exercise	43
Remark	43
6.2.1 Definition (Kernel and Image of a Homomorphism)	43
6.2.2 Proposition	43
6.2.3 Exercise	44
6.2.4 Proposition	44
6.2.5 Exercise	44
6.3.1 Definition (Isomorphisms)	45
Remark	45
6.3.2 Definition (Group Isomorphisms)	45
Remark	45
6.3.3 Exercise	45
6.3.4 Theorem	45
6.3.5 Exercise	46
6.4.1 Theorem (The First Isomorphism Theorem (early version))	46
6.4.2 Definition (Induced Isomorphisms)	47
6.4.3 Theorem (The First Isomorphism Theorem)	47
6.4.4 Exercise	48
6.5.1 Theorem	48
6.5.2 Theorem	48
7.1.1 Definition (Quotient Set)	49
Remark	49
7.1.2 Exercise	49
Remark	49
7.1.3 Definition (Normal Subgroups)	50
Remark	50
7.1.4 Proposition	50
Remark	50
7.1.5 Definition (Quotient Groups)	50
Remark	50
Remark	51
7.1.6 Exercise (Coset Operation Preserves Size)	51
7.1.7 Exercise	51
7.1.8 Proposition	52
7.1.9 Exercise	52
Remark	52
7.1.10 Exercise	52
7.1.11 Exercise	52
7.1.12 Theorem	52
Remark	53
7.2.1 Exercise	53
7.2.2 Theorem	53
Remark	53
7.2.3 Definition (Product of Group Subsets)	54
Remark	54
7.2.4 Exercise	54
7.2.5 Lemma (Coset Construction)	54

Remark	54
7.2.6 Lemma	55
7.2.7 Theorem (Coset Operation)	55
Remark	55
7.2.8 Proposition (Simplified Coset Operation)	56
7.2.9 Corollary	56
7.2.10 Exercise	56
Remark	56
7.2.11 Exercise (Coset Product Preserves Size)	56
7.3.1 Corollary	57
7.3.2 Theorem (The Correspondence Theorem (Lattice Isomorphism Theorem))	57
Remark	57
Remark	59
8.1.1 Definition (Group Actions)	60
Remark	60
8.1.2 Definition (Orbits and Stabilizers)	60
8.1.3 Exercise (Orbits Are Equally Sized)	60
8.1.4 Theorem	60
Remark	60
8.1.5 Theorem (Orbit-stabilizer Theorem)	61
8.1.6 Proposition	61
Remark	61
8.1.7 Theorem (Cauchy's Theorem)	62
8.2.1 Definition (Fixes)	62
8.2.2 Proposition	62
8.2.3 Lemma (Burnside's Lemma)	62
A.1.1 Theorem (Remainder Theorem)	64
A.2.1 Corollary	64
A.2.2 Lemma	64
A.2.3 Lemma	64
A.2.4 Lemma	64
A.3.1 Lemma (Bezout's Lemma)	65
A.3.2 Theorem (Solvability of LDE)	65
B.1.1 Definition (Injective Functions)	66
B.1.2 Exercise	66
B.1.3 Definition (Surjective Functions)	66
B.1.4 Exercise	66
B.2.1 Definition (Bijections)	66
B.2.2 Corollary	66
B.2.3 Lemma (Invertibility of Bijections)	67
B.2.4 Exercise	67
B.2.5 Exercise	68
B.2.6 Exercise (Group Bijection)	68
B.2.7 Exercise	68
B.2.8 Exercise	68
B.2.9 Exercise	68
B.2.10 Exercise (Closure of Bijection Composition)	68
B.2.11 Exercise (Associativity of Bijection Composition)	69